



Decentralized Crypto World Bank

A Blockchain-Based Lending Platform
with AI-Enhanced Security

by

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Declaration

It is hereby declared that

1. The report submitted is our own original work while completing degree at BRAC University.
2. The report does not contain material previously published or written by a third party, except where this is appropriately cited through full and accurate referencing.
3. The report does not contain material which has been accepted, or submitted, for any other degree or diploma at a university or other institution.
4. We have acknowledged all main sources of help.

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Ethics Statement

This project operates exclusively on public blockchain test networks using test tokens with no real monetary value. No real financial transactions, personal banking data, or user financial records are involved at any stage of development or evaluation. All transaction data used for Artificial Intelligence and Machine Learning (AI/ML) model training and evaluation is either synthetically generated or sourced from publicly available anonymized datasets. The platform prototype does not process Know Your Customer (KYC) or Anti-Money Laundering (AML) data in its current scope, and all wallet addresses used during testing are disposable testnet addresses. We acknowledge the ethical considerations inherent in AI-assisted financial decision-making and have incorporated explainability mechanisms (SHapley Additive exPlanations, or SHAP) to ensure that automated risk assessments remain transparent and auditable by human reviewers.

Abstract

Global development finance relies on multilayered institutional structures to distribute capital across borders and communities, yet these systems often struggle with fragmented transparency, procedural friction, and uneven access to credit. Although decentralized finance has demonstrated that programmable infrastructures can automate lending and enhance auditability, existing models largely employ flat architectures that do not reflect institutional hierarchies or structured governance. As a result, there remains limited exploration of how tiered financial systems might be represented within decentralized environments while preserving oversight and adaptability. Here we present *Crypto World Bank*, a prototype framework that models a multi-tier institutional lending architecture on a programmable blockchain infrastructure. We show that hierarchical capital flows, role-based governance, and data-informed risk analytics can be coordinated within a unified smart contract environment, enabling transparent state visibility alongside adaptive decision support. This hybrid design illustrates how institutional finance and decentralized systems may converge, suggesting a pathway toward more accountable, flexible, and inclusive digital credit ecosystems.

Keywords: Blockchain, Decentralized Finance, Institutional Architecture, Financial Inclusion, Smart Contracts, AI-Augmented Governance

Dedication

Dedicated to our families for their unwavering support throughout our academic journey.

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Chapter 1

Introduction

The convergence of blockchain technology, decentralized finance, and artificial intelligence presents new opportunities to address longstanding inefficiencies in global development finance. This project, the Crypto World Bank, explores how a decentralized lending platform with a multi-tier institutional structure can improve transparency, reduce settlement friction, and expand access to credit in underserved markets. The complete source code, smart contracts, and supplementary documentation for this project are available in our [GitHub repository](#).

Cross-border finance remains constrained by slow correspondent-banking settlement, fragmented ledger visibility, and high transfer costs. International remittances reached approximately \$860 billion annually in 2024, yet \$48–56 billion is still lost to transfer fees [26]. At the same time, approximately 1.4 billion adults remain unbanked globally [14], and the financing gap for micro, small, and medium enterprises exceeds \$4.5 trillion [20]. In parallel, decentralized finance has shown that lending logic can be automated transparently on blockchain infrastructure, but leading protocols continue to employ flat, pool-based architectures that do not model real-world institutional hierarchies or the directional capital flows observed in development finance.

We position blockchain in this context as a coordination technology that provides shared state visibility, tamper-evident audit trails, and programmable process enforcement across multiple institutions. The platform further addresses governance design, security controls, and regulatory considerations through a phased deployment pathway involving academic validation and regulatory sandbox pilots. By integrating a four-tier institutional hierarchy with transparent ledger infrastructure and a lightweight AI-assisted monitoring layer, the Crypto World Bank explores a hybrid architecture for institutional decentralized finance that targets improved operational efficiency and expanded access to credit in emerging markets.

1.1 Background

International development finance is delivered through layered institutional structures in which multilateral development institutions (e.g., World Bank, International Monetary Fund), national financial intermediaries, and local lenders jointly channel capital to

end borrowers. While this model enables risk distribution and local market access, it is associated with significant operational challenges. Cross-border correspondent banking requires financial institutions to maintain pre-funded nostro and vostro accounts at correspondent banks in each currency corridor, trapping substantial capital in idle balances that cannot be deployed for lending or investment [24]. Settlement of cross-border transactions routinely takes two to five business days, with average fees of approximately \$42 per transaction through the correspondent banking network [25]. The global remittance market—valued at approximately \$860 billion annually—loses an estimated \$48–56 billion per year to transfer fees, with the global average cost standing at 6.49% as of 2024, more than double the United Nations Sustainable Development Goal target of 3% [26].

Simultaneously, approximately 1.4 billion adults remain unbanked globally according to the World Bank Global Findex [14], with the majority concentrated in developing economies where documentation barriers, geographic remoteness, and high minimum balance requirements exclude large populations from formal financial services. The International Finance Corporation estimates the global financing gap for micro, small, and medium enterprises at \$4.5 trillion annually [20], representing unmet credit demand that existing institutions cannot efficiently serve through traditional intermediation chains.

The DeFi ecosystem has demonstrated that smart contract platforms can automate lending, collateral management, and interest computation with full on-chain transparency. Aave v3, the largest DeFi lending protocol, has accumulated over \$26 billion in total value locked across ten blockchain networks [27], while the broader DeFi lending market has surpassed \$55 billion in TVL [13]. However, these protocols employ flat, pool-based architectures in which all lenders deposit into a single pool and all borrowers draw from the same source—regardless of institutional role, creditworthiness, or geographic context. No existing DeFi protocol models the multi-tier institutional hierarchy, cross-tier capital flows, or tiered borrower access rules that characterize real-world development finance.

1.2 Rationale of the Study

This study is motivated by the convergence of three factors: (1) the persistent inefficiencies in traditional development finance that limit transparency and settlement speed; (2) the gap between DeFi’s technical capabilities and its failure to address multi-tier institutional lending use cases; and (3) the opportunity to combine blockchain’s auditability with lightweight analytics and monitoring support to improve operational oversight. By designing a prototype that bridges these domains, we aim to demonstrate a viable path toward transparent, programmable, and institutionally meaningful lending

for underserved populations.

1.3 Problem Statement

The global development finance ecosystem is characterized by a hierarchical topology in which supranational entities (e.g., the World Bank, International Monetary Fund) allocate capital to sovereign national institutions, which in turn disburse funds to regional and local banking entities, ultimately reaching end-user borrowers. This multi-layered intermediation introduces several systemic inefficiencies:

1. **Opacity of ledger state.** Internal bookkeeping across tiers is not publicly auditable. Stakeholders at lower tiers have limited visibility into reserve adequacy, approval criteria, and fund utilization at higher tiers. Traditional bank reserve ratios are self-reported and audited quarterly at best, providing no real-time solvency verification.
2. **Settlement latency and trapped capital.** Cross-border capital transfers between tiers involve multiple intermediaries, compliance checkpoints, and reconciliation processes, resulting in settlement periods ranging from two to five business days. The correspondent banking system requires banks to maintain pre-funded accounts (nostro/vostro) at partner institutions, with each currency corridor requiring a separate balance. This structural requirement traps significant capital in idle positions that cannot be deployed productively [24]. Transaction costs in the correspondent banking network average approximately \$42 per transfer [25], creating a cost barrier that disproportionately affects smaller transactions and developing-economy participants.
3. **Manual risk assessment.** Creditworthiness evaluation and fraud detection are predominantly human-driven, introducing inconsistency, bias, and scalability constraints. The absence of portable, on-chain credit histories forces repeated assessment at each institutional tier, adding friction and cost to the lending chain.
4. **Trust deficit and exclusionary access.** The effort required to establish inter-institutional trust through legal agreements, audit engagements, and compliance certifications constitutes a significant overhead that disproportionately affects smaller institutions and borrowers in developing economies. The World Inequality Report 2026 documents that the global monetary system systematically transfers resources from developing to developed economies, with developing nations recording negative return differentials averaging minus three percentage points compared to developed economies [28]. This structural disadvantage compounds the difficulty of accessing international credit markets.

Simultaneously, the DeFi ecosystem has demonstrated that smart contract platforms

can automate lending, collateral management, and interest computation with full transparency. Aave v3 has accumulated \$26.3 billion in TVL with \$17.7 billion in active borrows [27]; Compound v3 manages \$1.4 billion in TVL [29]; MakerDAO (now rebranded as Sky Protocol) issues approximately \$10.5 billion in combined DAI and USDS stablecoins with projected 2026 revenue of \$611 million [30]. Despite this scale, these protocols exhibit fundamental limitations in the context of institutional development finance:

- They employ **flat, peer-to-pool** or **over-collateralized** lending models in which all participants—from individual retail users to large institutional borrowers—interact with the same undifferentiated liquidity pool. No existing protocol models institutional hierarchy, tiered capital distribution, or cross-tier lending flows analogous to the World Bank → National Bank → Local Bank structure of development finance.
- They lack integration with **AI-driven risk analytics** and **explainable decision support** systems. DeFi lending decisions are governed entirely by collateral ratios and utilization curves, with no behavioral fraud detection, anomaly identification, or explainable credit assessment.
- They do not model **role-based, tiered governance** analogous to development finance institutions. Governance is exercised through token-weighted voting in which large token holders dominate decision-making, without the institutional role differentiation (regulator, wholesale lender, retail lender, borrower) required for hierarchical finance.
- Emerging institutional DeFi protocols such as Maple Finance (\$2.6–3.8 billion TVL, undercollateralized institutional lending) [31] and Goldfinch (\$680 million in loan originations across 18+ developing countries) [32] have introduced credit-based and emerging-market lending but still operate as single-tier platforms without hierarchical capital flows or interbank lending mechanics.

1.4 Objectives

The objectives of this project are:

1. To design and implement a four-tier lending architecture (World Bank → National Bank → Local Bank → Borrower) on an EVM-compatible blockchain that preserves institutional hierarchy while enabling shared ledger visibility.
2. To explore a lightweight off-chain analytics support layer—including fraud monitoring, anomaly identification, and explainable review support—to augment human credit governance.
3. To demonstrate transparent, programmable lending workflows with configurable

borrowing limits, installment schedules, and role-based access control enforced by smart contracts.

4. To validate the prototype on public testnets (Polygon, Ethereum Sepolia) with functional verification of hierarchical controls, transparency properties, and operational workflows.
5. To document governance design, security controls, regulatory considerations, and a phased deployment pathway for academic validation and potential regulatory sandbox pilots.

1.5 Research Questions

The study is guided by the following research questions:

1. **RQ1:** Can a hierarchical blockchain-based lending architecture model the real-world capital flow structure of development finance more faithfully than existing flat DeFi lending protocols?
2. **RQ2:** Can on-chain transparency, programmable reserve controls, and role-based governance reduce settlement opacity and coordination friction across multiple institutional tiers?
3. **RQ3:** Can a lightweight off-chain analytics layer provide practical and auditable support for fraud monitoring and lending oversight in such a system?
4. **RQ4:** Is the proposed architecture technically and operationally feasible on current public test networks while remaining relevant to real-world institutional and financial-inclusion use cases?

1.6 Blockchain Justification

The problem described above is fundamentally a **multi-party coordination and trust challenge** involving disparate institutions with potentially misaligned incentives. We assert that blockchain technology addresses this class of problems more effectively than any conventional alternative for the following reasons:

1. **Trust minimization through consensus.** Distributed consensus mechanisms ensure that no single party can unilaterally alter the ledger state, eliminating the need for a trusted central operator.
2. **Programmable enforcement.** Smart contracts codify lending rules—borrowing limits, installment schedules, approval workflows—as deterministic, self-executing programs, reducing reliance on manual processes and bilateral agreements.
3. **Cryptographic auditability.** All state transitions are recorded in an immutable, publicly verifiable transaction log, enabling real-time auditing by regulators, partners, and borrowers without requiring costly third-party engagements.

4. **Composable incentive structures.** On-chain reputation systems, governance tokens, and programmable fee distributions can align incentives across all network participants.

A conventional cloud-hosted database infrastructure would necessitate a trusted central operator to maintain ledger integrity, enforce access control, and provide audit guarantees—reintroducing the very trust dependencies that this project seeks to mitigate. Blockchain eliminates this single point of trust failure.

1.7 Proposed Solution

The Crypto World Bank implements a **four-tier, on-chain lending architecture**:

- **Tier 1 — World Bank:** Maintains the global crypto reserve. Allocates capital to registered National Banks through smart contract-mediated lending operations.
- **Tier 2 — National Banks:** Borrow from the World Bank reserve. Lend to registered Local Banks within their jurisdiction. Aggregate risk exposure at the national level.
- **Tier 3 — Local Banks:** Borrow from National Banks. Process loan requests from individual borrowers. Employ designated approvers for loan lifecycle management.
- **Tier 4 — Borrowers:** Submit loan requests to Local Banks. Repay through configurable installment schedules. Accumulate on-chain repayment history that informs future borrowing limits.

The platform further integrates:

- **Risk-aware borrowing limits** computed from rolling 6-month and 1-year transaction windows.
- **Automatic installment generation** for loan amounts exceeding a configurable threshold (e.g., 100 ETH equivalent).
- **Off-chain AI/ML security analytics** for fraud detection (Random Forest), anomaly identification (Isolation Forest), and explainable risk assessment (SHAP).

1.7.1 Cross-Tier Lending System

In traditional banking, capital does not flow strictly top-down. Banks at the same level routinely lend to each other through the interbank lending market (e.g., the federal funds market), and lower-tier banks with surplus reserves deposit excess capital upward with their parent institutions or central banks. The Crypto World Bank extends its hierarchical architecture to support these **multi-directional lending flows**, creating a more realistic and resilient financial ecosystem.

Same-Tier Lending

Banks at the same hierarchical level can lend to one another to manage short-term liquidity:

- **National Bank ↔ National Bank:** A national bank with excess reserves can lend to another experiencing a liquidity shortfall, analogous to the traditional interbank lending market. The lending rate floats based on supply and demand across national banks in the system, similar to the Secured Overnight Financing Rate (SOFR).
- **Local Bank ↔ Local Bank:** Local banks within the same or different national jurisdictions can share liquidity through a peer lending pool. This prevents localized liquidity crunches when one bank has excess deposits while another faces elevated loan demand.

Same-tier lending is implemented through an on-chain **InterBankLendingPool** contract at each tier level, where banks with surplus offer short-term liquidity and banks with deficit can borrow at utilization-based floating rates.

Upward Lending

Lower-tier banks with surplus capital can lend upward through the hierarchy:

- **Local Banks → National Banks:** When local banks accumulate reserves beyond their minimum reserve ratio, they can deposit surplus capital into their parent national bank's liquidity pool, earning a deposit yield. This mirrors how commercial banks deposit excess reserves with central banks through standing deposit facilities.
- **National Banks → World Bank:** National banks can contribute surplus to the global reserve, strengthening the system's overall capital base. This provides a return channel for well-capitalized national banks and increases the reserve available for downward distribution to banks with higher demand.

Upward lending rates are lower than downward lending rates (reflecting the lower risk of lending to a higher-tier institution), creating a natural interest rate term structure across the hierarchy.

Tiered Borrower Access

Different borrower classes access different tiers based on loan size and organizational type:

Table 1.1: Borrower tier access rules.

Borrower Type	Accessible Tiers	Loan Range	Use Case
Individual / End User	Local Bank only	0.01–10 ETH	Personal, micro-enterprise
Small Business / SME	Local Bank, National Bank	1–100 ETH	Working capital, equipment
Large Corporate	National Bank, World Bank	50–10,000 ETH	Infrastructure, large projects
Institutional / Sovereign	World Bank only	1,000+ ETH	Development programs

End users and retail borrowers access the system exclusively through Local Banks, preserving the hierarchical intermediation model. Large corporate accounts and institutional borrowers can access higher tiers directly, subject to enhanced verification (on-chain credit history combined with off-chain documentation) and a borrower class designation in the smart contract that determines tier access permissions.

This multi-directional lending model—combining downward capital distribution, same-tier interbank lending, upward surplus repatriation, and tiered borrower access—creates a more complete representation of real-world banking flows within the decentralized architecture.

1.8 Methodology in Brief

We adopted a lightweight Agile/Scrum methodology with three sprints over an 8-week development window. Sprint 1 establishes smart contracts, frontend scaffolding, wallet integration, and database schema. Sprint 2 delivers lending features (loan requests, approvals, installments, borrowing limits), chat system, income verification, and bank registration. Sprint 3 integrates AI/ML fraud detection, risk dashboard, security audit, and documentation. Development uses Solidity 0.8.20 for smart contracts, React 18 with TypeScript for the frontend, FastAPI for the backend, and scikit-learn for ML models. We deploy on Polygon and Ethereum testnets at zero real-cryptocurrency cost.

1.9 Scopes and Challenges

Scope. Our prototype operates exclusively on public testnets (no real cryptocurrency). In-scope features include: four-tier lending, loan request and approval workflow, install-

ment payments, borrowing limit computation, borrower–bank chat, income verification, fraud detection (Random Forest + SHAP), and anomaly detection (Isolation Forest). The dual-currency facility is designed as an auxiliary add-on to existing banking infrastructure. Out of scope for the prototype: mainnet deployment, fiat on-ramp/off-ramp integration, KYC/AML compliance automation, and production-scale stress testing.

Challenges. Key challenges include: (1) *Data scarcity*—labeled fraud data for DeFi lending is limited; synthetic or transfer learning may be required. (2) *Regulatory uncertainty*—crypto lending faces jurisdictional restrictions; our prototype mitigates this by operating on testnets only. (3) *Gas cost sensitivity*—complex on-chain logic increases transaction cost; the design partitions AI/ML and heavy computation off-chain. (4) *Model interpretability*—regulatory requirements for explainable lending decisions are addressed via SHAP-based feature attribution. (5) *Economic sustainability*—the platform must generate sufficient interest income to offset gas costs incurred across all four tiers; this balance is analyzed in Chapter 5.

1.9.1 Economic Sustainability: Interest Revenue Versus Gas Costs

A critical concern for any blockchain-based lending platform is whether the interest revenue generated from lending operations can sustainably exceed the cumulative gas costs of on-chain transactions. Each tier in the hierarchy incurs gas fees for deposits, loan requests, approvals, disbursements, repayments, and installment processing. On Ethereum mainnet, a single complex smart contract interaction can cost \$5–50 depending on network congestion; however, on Polygon PoS, the same operation costs approximately \$0.001–0.01. At an average retail loan of 10 ETH (\$20,000 at \$2,000/ETH) with an 8% APR, the borrower pays \$1,600 in annual interest. Even with 10–15 on-chain transactions per loan lifecycle (request, approval, disbursement, 12 monthly installments), total gas costs on Polygon remain below \$0.15—representing less than 0.01% of the interest generated. The platform therefore achieves a sustainable margin on Layer 2 networks, though mainnet deployment would require careful gas optimization or batched transaction processing to maintain viability for micro-loans.

1.9.2 Resistance to Institutional Capture

If the Crypto World Bank achieves widespread adoption, a legitimate concern is whether large capitalists and incumbent financial institutions—who currently control traditional banking—could migrate to this system and replicate existing patterns of financial concentration. The platform addresses this risk through several architectural safeguards:

- **Transparent reserve ratios:** Unlike traditional banks where reserve adequacy is self-reported and audited quarterly, on-chain reserves are publicly verifiable

in real time. No institution can obscure its true financial position to gain unfair advantage.

- **Algorithmic interest rates:** Interest rates are bounded by smart contract parameters rather than set through opaque internal pricing committees, preventing exploitative rate-setting.
- **Immutable audit trails:** Every transaction, approval, and governance decision is permanently recorded on-chain, making regulatory capture and corruption detectable by any participant.
- **Open-source governance:** The smart contract code is publicly auditable, preventing hidden backdoors or preferential treatment encoded at the protocol level.
- **Tiered access with caps:** Borrowing limits and tier access rules are enforced programmatically, preventing any single entity from monopolizing the capital pool.

While no system can fully prevent wealth concentration in a free market, the transparency and programmable enforcement properties of blockchain infrastructure raise the cost and detectability of exploitative behavior substantially compared to traditional opaque intermediaries.

1.9.3 Cryptocurrency Supply Constraints and Scalability

A fundamental question arises regarding the use of fixed-supply cryptocurrencies (such as Bitcoin's 21 million cap or Ethereum's deflationary issuance post-Merge) as the denomination for a global lending system: what happens when demand for credit exceeds the available supply of the underlying cryptocurrency?

This concern is addressed through three design considerations:

1. **Stablecoin integration (planned):** The platform's roadmap includes ERC-20 stablecoin support (USDC, USDT, DAI), which are pegged to fiat currencies and can be minted in response to demand—providing elastic supply without abandoning on-chain transparency.
2. **Layer 2 and multi-chain deployment:** Cross-chain interoperability enables the platform to operate across multiple networks simultaneously, distributing demand across different token ecosystems and preventing single-chain supply bottlenecks.
3. **Fractional reserve design:** The hierarchical lending model inherently operates as a fractional reserve system—each ETH deposited at Tier 1 can generate multiple loans downstream through the multiplier effect, just as in traditional banking. The total lending capacity of the system exceeds the raw token supply by a factor determined by the reserve ratio at each tier.

The fixed supply of the underlying cryptocurrency is not a limitation but a feature: it prevents the discretionary monetary expansion that erodes purchasing power in traditional fiat systems. For scalability, the platform relies on stablecoins and multi-chain deployment rather than inflating the base currency—preserving the deflationary monetary discipline that distinguishes decentralized finance from traditional central banking.

1.10 Market Analysis and Partnership Ecosystem

1.10.1 Market Sizing

Table 1.2: Market segments.

Segment	Description	Estimated Scale
Total Addressable Market (TAM)	Global DeFi lending (\$55B+ TVL [13]); cross-border remittances (\$860B [26]); SME gap (\$4.5T [20])	\$55B–5T+
Serviceable Addressable Market (SAM)	Institutional and semi-institutional lending requiring hierarchical structures	\$5–15B
Serviceable Obtainable Market (SOM)	Pilot deployments in regulatory sandboxes, academic prototypes, NGO-backed microfinance	\$50–200M

Sources: (a) TAM: DefiLlama [13] reports DeFi lending TVL exceeding \$55B; World Bank [26] values global remittances at \$860B; IFC [20] estimates the MSME financing gap at \$4.5T; (b) SAM and SOM: project-derived estimates based on industry segmentation [18].

The market for **transparent, hierarchy-preserving decentralized lending** is presently unaddressed by existing DeFi protocols, which universally adopt flat, pool-based architectures. This represents a significant whitespace opportunity.

1.10.2 Target Customer Segment

The Crypto World Bank targets the **retail customer segment** — individual borrowers and small businesses seeking transparent, accessible crypto-based lending services.

Table 1.3: Target customer segment profile.

Characteristic	Description
Primary Users	Individual retail borrowers seeking personal or small business loans
Geographic Focus	Developing economies with limited traditional banking access (e.g., Bangladesh, Southeast Asia, Sub-Saharan Africa)
Loan Size Range	Micro to mid-range: 0.1 ETH – 500 ETH equivalent (~\$200 – \$1,000,000 at current rates)
User Profile	Digitally literate individuals with cryptocurrency wallet access; small business owners; gig-economy freelancers
Key Pain Points	High interest rates from informal lenders; lack of credit history in traditional systems; exclusion from banking due to documentation barriers

Why Retail: The retail segment represents the largest underserved population in developing economies. According to the World Bank’s Global Findex Database, approximately 1.4 billion adults remain unbanked, with the majority concentrated in developing countries. By targeting retail customers, the platform maximizes financial inclusion impact while generating the transaction volume necessary to sustain the lending ecosystem. The hierarchical banking model (World Bank → National → Local → Borrower) mirrors how traditional microfinance institutions reach retail customers in underserved regions, but with blockchain-enforced transparency and AI-enhanced risk assessment.

1.10.3 Partner Ecosystem

Table 1.4: Partner categories and roles.

Partner Category	Functional Role	Blockchain-Mediated Incentive
Financial Regulators	Regulatory sandbox approval; compliance oversight	Reduced enforcement cost through on-chain transparency and audit trails
Banking Institutions	Network membership as National/Local Banks	Access to diversified global reserve; reduced inter-bank settlement friction
Payment Gateway Providers	Fiat-to-crypto on-ramp and off-ramp services	Volume-based transaction fees; expanded market reach
Academic & Research Institutions	Validation of AI/ML models; publication of research findings	Access to anonymized datasets; collaborative research opportunities
Non-Governmental Organizations	Pilot deployment; field testing with underserved borrower populations	Transparent, low-friction credit access for beneficiaries

1.10.4 Incentive Alignment Through the Blockchain Platform

Partner incentives are allocated and enforced through the blockchain platform itself:

- **Immutable repayment records** serve as on-chain reputation signals, enabling data-driven credit decisions without third-party credit bureaus.
- **Transparent reserve verification** allows partners to independently confirm that allocated capital is deployed as committed.
- **Programmable fee structures** (future extension) distribute transaction fees proportionally to network participants based on their contribution to lending volume and governance.

Chapter 2

Literature Review

2.1 Preliminaries

This section establishes key terminology and concepts used throughout the literature review and the project design.

Decentralized Finance (DeFi). DeFi refers to financial services built on blockchain platforms that operate without traditional intermediaries. Lending protocols enable users to borrow and lend assets through smart contracts, with interest rates typically determined algorithmically or by market utilization.

Smart contracts. Self-executing programs deployed on a blockchain that execute when predefined conditions are met. In lending, smart contracts handle deposits, loan requests, approvals, repayments, and interest computation without human intermediaries.

Over-collateralization. A lending model where borrowers must pledge collateral worth more than the loan amount. This is the dominant model in DeFi (e.g., Aave, Compound) due to the absence of credit scoring; it excludes under-collateralized lending common in traditional finance.

Explainable AI (XAI). Techniques that make machine learning model predictions interpretable to humans. SHAP and Local Interpretable Model-agnostic Explanations (LIME) are prominent methods for attributing feature importance to individual predictions, critical for regulatory compliance in lending decisions.

Proof of Stake (PoS). A consensus mechanism where validators secure the network by staking tokens; used by Ethereum (post-Merge) and Polygon. PoS is more energy-efficient than Proof of Work and enables faster block finality.

Central Bank Digital Currency (CBDC). A digital form of a country's fiat currency issued by the central bank. CBDCs use tiered distribution models that parallel the multi-tier architecture explored in this project.

2.2 Review of Existing Research

The design of the Crypto World Bank draws on a review of peer-reviewed publications, institutional reports, and industry data spanning multiple domains directly relevant to our architecture: decentralized lending protocol design, machine learning for blockchain security, explainable AI in financial systems, blockchain for financial inclusion, smart contract security, correspondent banking economics, monetary policy distribution effects, and real-world asset tokenization.

2.2.1 Decentralized Lending and DeFi Protocol Design

Werner et al. [1] systematize DeFi protocol design in their SoK paper, identifying that existing lending platforms are uniformly pool-based and over-collateralized, lacking institutional hierarchy—a gap this project directly addresses. Their taxonomy of DeFi protocols reveals that no existing system models multi-tier capital flow analogous to development finance, confirming the novelty of our four-tier approach.

Bastankhah et al. [2] introduce an adaptive, data-driven DeFi lending protocol with a dual fast/slow control architecture, demonstrating that dynamic interest rate adjustment significantly outperforms static utilization curves used by Aave and Compound. Their work validates the feasibility of algorithmic lending parameter optimization, an approach we extend through our configurable interest rate tiers across the four-level hierarchy.

Li et al. [46] present a multi-chain lending model that distributes lending operations across multiple blockchain networks to improve throughput and reduce single-chain congestion. Their cross-chain settlement mechanism validates the technical viability of operating lending protocols across heterogeneous blockchain environments, directly supporting our planned multi-chain deployment strategy across Ethereum and Polygon.

Xu et al. [47] develop an evaluation system for DeFi lending protocols that quantifies protocol risk through metrics including utilization ratio stability, liquidation efficiency, and interest rate responsiveness. Their evaluation framework provides a benchmark methodology that can be adapted to assess the performance of our hierarchical lending tiers against existing flat-pool protocols.

Sharma et al. [48] propose a blockchain-empowered framework for peer-to-peer lending that eliminates intermediary overhead through smart contract-mediated loan origination and repayment. While their architecture remains single-tier, their analysis of gas cost optimization strategies and borrower identity management through on-chain credentials informs our approach to minimizing transaction costs across the four-tier hierarchy.

2.2.2 Machine Learning for Blockchain Security

Palaiokrassas et al. [3] apply machine learning to multichain DeFi fraud detection across 23 protocols encompassing over 54 million transactions, demonstrating that DeFi-specific behavioral features improve XGBoost and Neural Network classifiers to F1-scores of 0.76–0.85, compared to 0.08 with transactional features alone. Their finding that behavioral features outperform raw transaction data directly informed our feature engineering approach for the Random Forest fraud detection model.

Liu et al. [6] introduce the Isolation Forest algorithm for unsupervised anomaly detection. The algorithm isolates anomalies by recursively partitioning data, assigning shorter path lengths to outliers. We adopted Isolation Forest as our secondary detection model for wallet behavior analysis where labeled fraud data is scarce, enabling detection of novel attack patterns without pre-labeled training data.

Hassan et al. [49] combine blockchain and machine learning for fraud detection using a privacy-preserving and adaptive incentive-based approach. Their framework trains ML classifiers on encrypted transaction features using federated learning, ensuring that individual transaction data is never exposed to the detection model’s operators. This privacy-preserving paradigm is relevant to future extensions of our AI/ML layer, where borrower transaction histories must remain confidential while still enabling system-wide fraud pattern recognition.

2.2.3 Explainable AI in Financial Decision Systems

Adom et al. [4] provide a direct comparison of LIME and SHAP on loan approval systems, finding that SHAP provides deeper, more consistent feature attributions via Shapley values, while LIME offers faster runtime but less stable explanations across repeated evaluations. Their comparative analysis informed our decision to adopt SHAP as the primary explainability method for loan risk assessments, balancing interpretability depth with regulatory compliance requirements.

Gupta et al. [56] apply SHAP-based interpretable models to credit default assessment, demonstrating that SHAP feature attributions on Random Forest and Gradient Boosting classifiers achieve AUC scores above 0.92 while maintaining full interpretability of individual predictions. Their methodology for generating per-borrower risk explanations directly informs our planned credit risk dashboard, where loan approvers review SHAP waterfall plots before making lending decisions.

2.2.4 Blockchain for Financial Inclusion

Tan [5] develops an International Monetary Fund (IMF) model showing CBDCs in developing countries can bank large unbanked populations through a two-tier distribution

model (central bank $\rightarrow$ commercial banks $\rightarrow$ users) that directly parallels our four-tier hierarchy. The study demonstrates that digital currency infrastructure, when distributed through existing institutional channels, can reach populations excluded from traditional banking—supporting the Crypto World Bank’s mission of transparent, programmable lending for underserved populations.

Alam et al. [54] investigate the application of blockchain technology for the microcredit industry specifically in Bangladesh, demonstrating that smart contract-based loan management can reduce administrative overhead by up to 60% compared to manual microfinance processes. Their study of the Bangladeshi financial landscape—where approximately 40% of adults lack access to formal banking—directly validates the geographic and demographic relevance of the Crypto World Bank’s target market.

Islam et al. [53] analyze the design requirements and challenges of Central Bank Digital Currencies, identifying a two-tier retail CBDC model (central bank to commercial banks to end users) as the preferred distribution architecture. Their analysis of interoperability challenges between CBDC systems and existing payment infrastructure informs our design of the dual-currency facility that bridges fiat and cryptocurrency within participating banks.

2.2.5 Smart Contract Security and Governance

Atzei et al. [7] catalog Ethereum smart contract attack vectors including reentrancy, integer overflow, and access control vulnerabilities. Their taxonomy directly informed our use of OpenZeppelin’s ReentrancyGuard, Solidity 0.8.20 built-in overflow protection, and role-based access control modifiers. We adopted their recommended defense-in-depth strategy of combining security primitives with planned formal verification using static analysis tools (Slither) and symbolic execution (Mythril).

Wang et al. [50] present ContractWard, an automated vulnerability detection system that applies machine learning classifiers (Random Forest, SVM, k-NN) to identify six categories of smart contract vulnerabilities from bytecode features. ContractWard achieves F1-scores of 0.96 for reentrancy detection and 0.93 for access control vulnerabilities, demonstrating that ML-based security auditing can complement traditional static analysis tools in our planned security verification pipeline.

Liao et al. [51] introduce SoliAudit, which combines machine learning classification with fuzz testing for smart contract vulnerability assessment. Their hybrid approach—using ML to prioritize likely vulnerable code paths followed by targeted fuzz testing—reduces audit time by an estimated 70% compared to manual review. SoliAudit’s methodology informs our planned security audit strategy for the three-contract architecture.

So et al. [52] develop VERISMART, a highly precise safety verifier for Ethereum smart contracts that uses constraint-based abstract interpretation to detect arithmetic overflow, division-by-zero, and array out-of-bounds errors. VERISMART achieves 98.4% precision with zero false positives on benchmark datasets, making it a candidate tool for formal verification of our smart contract invariants (e.g., reserve ratio maintenance, cascading interest rate bounds).

2.2.6 AI Security Features for Blockchain Lending

Beyond the fraud detection and anomaly identification models already integrated into the platform, several additional AI-driven security features can strengthen the Crypto World Bank’s resilience against evolving threats:

- **Graph Neural Network (GNN) transaction analysis:** GNN-based models can analyze the transaction graph structure to detect coordinated fraud rings—clusters of wallets that collude to manipulate borrowing limits or create circular lending patterns. Research by Palaiokrassas et al. [3] demonstrates that graph-based features significantly outperform flat transaction features for DeFi fraud detection.
- **Federated learning for cross-bank fraud intelligence:** Hassan et al. [49] demonstrate that federated learning enables multiple institutions to collaboratively train fraud detection models without sharing raw transaction data. Deploying federated learning across the Crypto World Bank’s National and Local Banks would enable system-wide threat detection while preserving per-bank data privacy.
- **Reinforcement learning for adaptive interest rates:** RL agents can dynamically adjust interest rate parameters in response to changing market conditions, utilization rates, and risk profiles—reducing the lag inherent in manual parameter governance.
- **Real-time smart contract monitoring:** ML-based monitoring agents (informed by ContractWard [50] and SoliAudit [51]) can continuously analyze incoming transactions for patterns matching known exploit signatures, triggering the platform’s pause mechanism before significant losses occur.
- **Natural language processing for income verification:** NLP models can automate the extraction and validation of income information from uploaded documents, reducing the manual review burden on bank officers while maintaining verification accuracy.

2.2.7 Gas Cost Optimization and Layer 2 Scalability

Tolmach et al. [55] analyze optimal gas fee minimization strategies in DeFi, demonstrating that transaction batching, calldata compression, and strategic timing of on-chain operations can reduce gas expenditure by 30–50% on Ethereum mainnet. Their findings

validate our architectural decision to deploy on Polygon PoS (where fees are already sub-cent) while informing optimization strategies for potential Ethereum mainnet deployment. The study also identifies that complex multi-step DeFi operations (analogous to our four-tier loan lifecycle) benefit disproportionately from Layer 2 deployment due to the multiplicative gas savings across sequential contract calls.

2.2.8 Correspondent Banking and Cross-Border Settlement

The Bank for International Settlements [21] and the Financial Stability Board [33] have extensively documented the structural inefficiencies of the correspondent banking network. The traditional model requires banks to maintain pre-funded nostro accounts in each currency corridor, creating a global pool of idle capital that cannot be deployed for productive lending. Settlement through the SWIFT network and correspondent banks involves message routing through MT103 payment instructions, compliance screening at each intermediary, and reconciliation processes that extend settlement times to two–five business days. The World Bank’s Migration and Development Brief [26] reports that the average cost of sending \$200 across borders stands at 6.49%, with Sub-Saharan African corridors averaging above 8%. These findings motivate our design of on-chain settlement with near-instant finality and sub-cent transaction costs on Layer 2 networks.

2.2.9 Monetary Policy Distribution and Financial Inequality

Recent empirical research has documented the distributional effects of monetary policy on wealth inequality. A 2025 cross-country study spanning 49 nations (1999–2019) found that central bank asset purchase programs have a “discernible relationship” with increased wealth inequality, with effects on wealth distribution being “more pronounced and persistent” than effects on income inequality [34]. The Federal Reserve’s own 2025 analysis using U.S. metropolitan tax data confirmed that contractionary monetary policy disproportionately reduces earnings for lower-income workers [35]. These findings provide economic motivation for the Crypto World Bank’s design principles: transparent and algorithmically determined monetary parameters (interest rates, reserve ratios, supply rules) encoded in smart contracts rather than set through discretionary central bank decisions. The platform’s on-chain transparency ensures that all participants observe the same interest rates and reserve conditions, eliminating the informational asymmetry that enables the Cantillon Effect—the phenomenon whereby early recipients of newly created money benefit at the expense of later recipients who face higher prices [36].

2.2.10 Real-World Asset Tokenization and Institutional DeFi

The tokenization of real-world assets represents a growing bridge between traditional and decentralized finance. R3’s Corda platform reports over \$17 billion in tokenized

real-world assets on its permissioned ledger as of September 2025, with institutional participants including HSBC and Bank of America [37]. Centrifuge has deployed across eight blockchain networks with over \$1 billion in tokenized institutional fund products [38]. On the public chain side, the World Bank itself has adopted blockchain for fund disbursement tracking through its FundsChain initiative (built on Hyperledger Besu), piloted across 13 projects in 10 countries and scaling to 250 projects by mid-2026 [39]. These developments validate both the institutional appetite for blockchain-based financial infrastructure and the viability of hierarchical fund distribution models on distributed ledgers. The Crypto World Bank extends this institutional adoption trajectory by implementing not merely fund tracking but a complete hierarchical lending system with on-chain interest rates, reserves, and credit assessment.

2.3 Literature Review Summary

Table 2.1 summarizes the key reviewed works, their methodologies, key findings, and relevance to our project, following the literature table format recommended by Michigan State University Libraries [23].

Table 2.1: Literature review summary table.

Author / Year	Research Focus	Methodology	Key Findings	Relevance to Our Project
Werner et al. (2022) [1]	DeFi protocol systematization	SoK survey of 12+ DeFi protocol categories	Lending platforms are uniformly pool-based and over-collateralized; no institutional hierarchy exists	Identifies the core gap our four-tier architecture addresses
Bastankhah et al. (2023) [2]	Adaptive DeFi lending	Dual fast/slow control; simulation on historical data	Dynamic rate adjustment outperforms static utilization curves	Validates algorithmic lending parameter optimization

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Table 2.1: Literature review summary table (continued).

Author / Year	Research Focus	Methodology	Key Findings	Relevance to Our Project
Palaiokrassas et al. (2023) [3]	ML fraud detection in DeFi	XGBoost, NN classifiers on 54M+ multi-chain transactions	DeFi behavioral features yield F1: 0.76–0.85 vs. 0.08 with transaction features alone	Informs our feature engineering for Random Forest fraud model
Adom et al. (2022) [4]	XAI in loan approval	LIME vs. SHAP comparison on lending datasets	SHAP provides deeper, more consistent attributions; LIME is faster but less stable	Justifies SHAP as our primary explainability method
Tan (2023) [5]	CBDC and financial inclusion	IMF economic model for developing nations	Two-tier CBDC distribution banks large unbanked populations	Directly parallels our four-tier institutional hierarchy
Liu et al. (2008) [6]	Anomaly detection	Isolation Forest algorithm on synthetic and real datasets	Isolation via recursive partitioning; shorter paths for anomalies	Adopted as secondary unsupervised detection for wallet behavior
Atzei et al. (2017) [7]	Smart contract security	SoK of Ethereum attack vectors	Catalogs reentrancy, overflow, access control vulnerabilities	Directly informs our security primitives and planned formal verification

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Table 2.1: Literature review summary table (continued).

Author / Year	Research Focus	Methodology	Key Findings	Relevance to Our Project
BIS/FSB [21][33]	Cross-border settlement	Analysis of correspondent banking infrastructure	Settlement takes 2–5 days; avg. cost \$42/transfer; capital trapped in nostro accounts	Motivates on-chain settlement with near-instant finality
Beyer et al. (2025) [34]	Monetary policy and inequality	Cross-country analysis, 49 nations, 1999–2019	QE increases wealth inequality; effects more persistent than income effects	Motivates transparent, algorithmic monetary parameters
World Bank (2025) [39]	Blockchain for development finance	FundsChain pilot on Hyperledger Besu, 13 projects, 10 countries	Blockchain tracking reduces reporting from months to minutes	Validates institutional appetite for blockchain-based fund distribution
Li et al. (2024) [46]	Multi-chain DeFi lending	Cross-chain lending model design and implementation	Multi-chain distribution improves throughput and reduces congestion	Supports our planned multi-chain deployment strategy
Xu et al. (2023) [47]	DeFi lending protocol evaluation	Quantitative risk metrics for lending protocol assessment	Utilization ratio, liquidation efficiency, and rate responsiveness as key metrics	Provides benchmark methodology for evaluating our hierarchical tiers

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Table 2.1: Literature review summary table (continued).

Author / Year	Research Focus	Methodology	Key Findings	Relevance to Our Project
Sharma et al. (2021) [48]	Blockchain P2P lending	Smart contract-mediated loan origination framework	Eliminates intermediary overhead; gas cost optimization analysis	Informs gas cost minimization across four-tier hierarchy
Hassan et al. (2022) [49]	Privacy-preserving fraud detection	Federated learning on encrypted blockchain features	Privacy-preserving ML achieves comparable accuracy to centralized models	Informs future privacy-preserving fraud detection extensions
Wang et al. (2020) [50]	Smart contract vulnerability detection	ML classifiers on bytecode features	F1: 0.96 for reentrancy, 0.93 for access control detection	ML-based auditing complements our planned security verification
Liao et al. (2019) [51]	Smart contract audit automation	Hybrid ML classification + fuzz testing	Reduces audit time by 70% vs. manual review	Informs our security audit strategy for three-contract architecture
So et al. (2020) [52]	Smart contract formal verification	Constraint-based abstract interpretation	98.4% precision, zero false positives on benchmarks	Candidate tool for verifying smart contract invariants
Islam et al. (2024) [53]	CBDC design requirements	Analysis of retail CBDC distribution architecture	Two-tier model preferred; interoperability challenges identified	Informs dual-currency facility design

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Table 2.1: Literature review summary table (continued).

Author / Year	Research Focus	Methodology	Key Findings	Relevance to Our Project
Alam et al. (2021) [54]	Blockchain micro-credit in Bangladesh	Smart contract-based microfinance management	Reduces admin overhead by 60%; validates Bangladesh market	Directly validates our target geographic and demographic market
Tolmach et al. (2024) [55]	DeFi gas fee optimization	Transaction batching and calldata compression analysis	Gas savings of 30–50% through optimization strategies	Validates Layer 2 deployment and informs mainnet optimization
Gupta et al. (2024) [56]	SHAP-based credit default models	SHAP on Random Forest and Gradient Boosting classifiers	AUC above 0.92 with full interpretability	Informs borrower risk explanation methodology

2.4 Summary of Key Findings

The literature review yields the following consolidated findings that directly inform our design:

- **DeFi lending is structurally flat.** Existing protocols (Aave, Compound, MakerDAO) use pool-based or over-collateralized models with total value locked exceeding \$55 billion [13]. No protocol models institutional hierarchy analogous to development finance. CBDC research [5] confirms that tiered distribution is viable and supports financial inclusion.
- **Correspondent banking is structurally inefficient.** Cross-border settlement takes two to five days with average costs of \$42 per transaction [25]. The system traps significant capital in idle nostro/vostro accounts [24], while remittance fees consume an estimated \$48–56 billion annually [26]. On-chain settlement on Layer 2 networks reduces both latency (to seconds) and cost (to sub-cent levels).
- **ML improves fraud detection in blockchain contexts.** DeFi-specific behavioral features substantially outperform transactional features alone (F1: 0.76–0.85 vs. 0.08 [3]). Isolation Forest [6] enables unsupervised detection where labeled

data is scarce.

- **SHAP satisfies regulatory explainability.** SHAP provides more consistent feature attributions than LIME for loan approval systems [4], meeting prudential requirements for transparent automated decision-making.
- **Smart contract security requires layered defense.** Reentrancy, overflow, and access control vulnerabilities are well-documented [7]. ML-based vulnerability detection achieves F1-scores above 0.93 [50], and hybrid ML-fuzz approaches reduce audit time by 70% [51]. Formal verification tools such as VERISMART achieve 98.4% precision [52]. OpenZeppelin primitives combined with automated verification are recommended for production deployments.
- **Monetary policy creates distributional inequality.** Cross-country evidence [34] and Federal Reserve research [35] confirm that quantitative easing disproportionately benefits asset holders while lower-income workers bear the costs of both inflation and subsequent tightening. Transparent, algorithmic monetary parameters can reduce this informational asymmetry.
- **Institutional blockchain adoption is accelerating.** The World Bank's FundsChain [39], R3 Corda's \$17 billion in tokenized assets [37], and JPMorgan Kinexys's multi-billion-dollar daily settlement volume [40] demonstrate that financial institutions are actively adopting blockchain for settlement, fund tracking, and asset tokenization.
- **Blockchain enables financial inclusion.** Approximately 1.4 billion adults remain unbanked globally [14], with the World Economic Forum identifying DeFi as a "leapfrog technology" enabling populations to bypass traditional banking infrastructure [41]. Celo's MiniPay wallet has onboarded 14 million users across 60+ countries with sub-cent transaction fees [42]. Blockchain-based microcredit in Bangladesh demonstrates 60% reduction in administrative overhead [54].
- **Multi-chain lending improves scalability.** Cross-chain lending models [46] and Layer 2 gas optimization strategies [55] demonstrate that distributing lending operations across multiple networks reduces congestion and transaction costs by 30–50%.
- **Privacy-preserving ML enables cross-institutional fraud detection.** Federated learning approaches [49] allow multiple lending institutions to collaboratively train fraud detection models without exposing individual transaction data, addressing the tension between system-wide security and per-institution data privacy.

These findings justify our four-tier architecture, AI/ML integration (Random Forest, Isolation Forest, SHAP), cross-tier lending design, governance framework, and planned security verification pipeline.

Chapter 3

System Architecture and Design

3.1 High-Level Architecture

The system employs a **three-layer decentralized application architecture**:

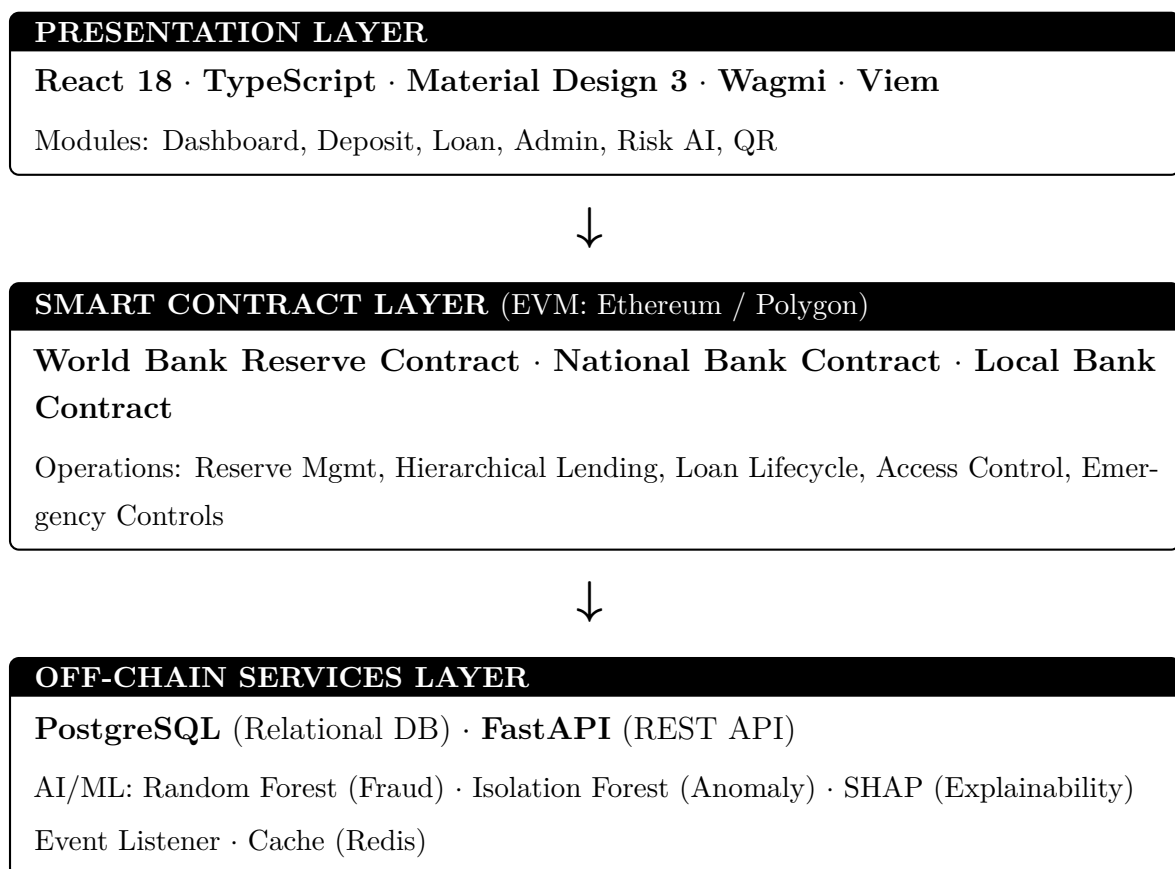


Figure 3.1 shows the component interactions across these three layers.

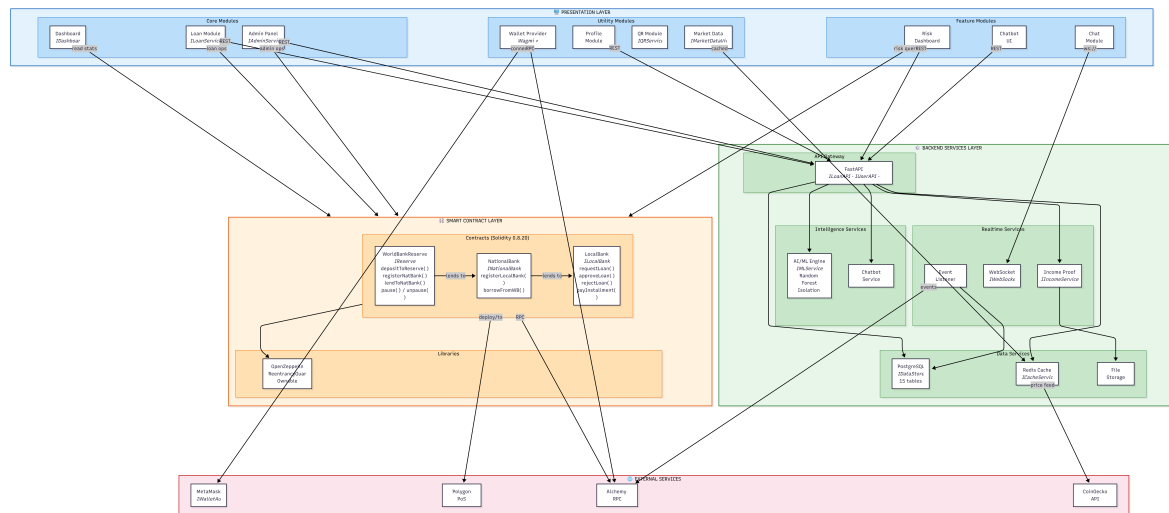


Figure 3.1: Component diagram showing interactions between the presentation layer, smart contract layer, off-chain backend services, and external systems.

3.2 Blockchain Platform Selection

Table 3.1: Blockchain platform selection criteria and justification.

Criterion	Selection	Justification
Platform	Ethereum Virtual Machine (EVM)	Largest developer ecosystem; battle-tested security model; extensive tooling
Network	Polygon Mumbai / Ethereum Sepolia (testnets)	Zero-cost deployment; production-equivalent behavior; free faucet access
Consensus	Proof-of-Stake (PoS) via Polygon validators	Energy-efficient; sub-2-second block finality; decentralized validator set
Smart contract language	Solidity 0.8.20	Industry standard; mature compiler with overflow protection; rich library ecosystem

3.2.1 Transaction Verification and Consensus

- **Polygon PoS:** Transactions are validated by a decentralized set of PoS validators. Block finality is achieved within approximately 2 seconds. Checkpoints are periodically committed to Ethereum mainnet for additional security.

- **On prototype testnets:** Mumbai and Sepolia employ analogous consensus models with no financial cost, enabling iterative development and testing.

3.3 Data Model and Database Design

The relational database schema comprises 15 normalized entities in Third Normal Form (3NF). Figure 3.2 presents the core system graph showing entity relationships, Figure 3.3 presents the full Entity-Relationship Diagram (ERD), and Figure 3.4 presents the Enhanced Entity-Relationship (EER) diagram.

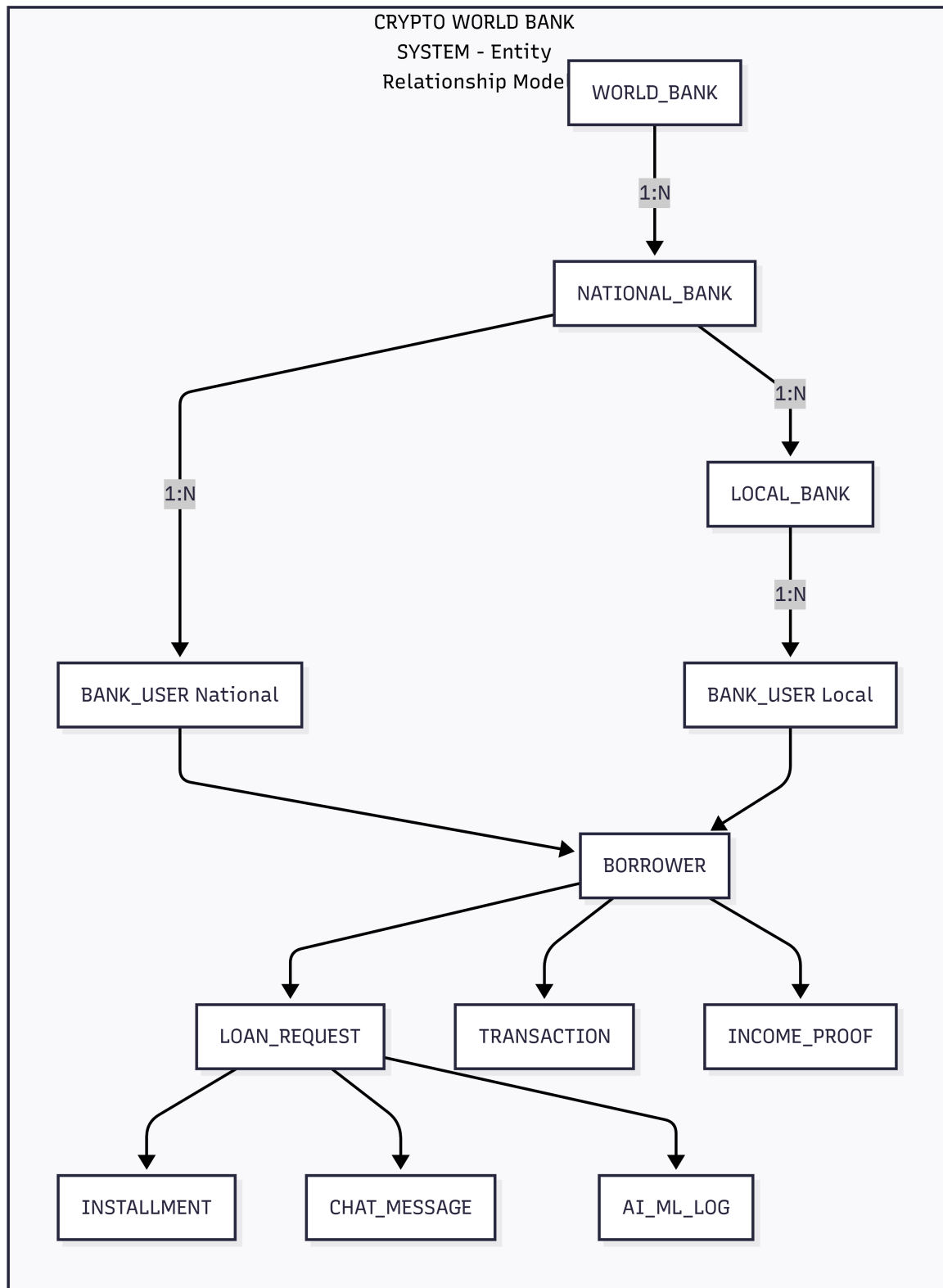


Figure 3.2: Core system graph showing the Crypto World Bank entity relationship model with hierarchical banking structure, lending lifecycle, communication, and security entities.

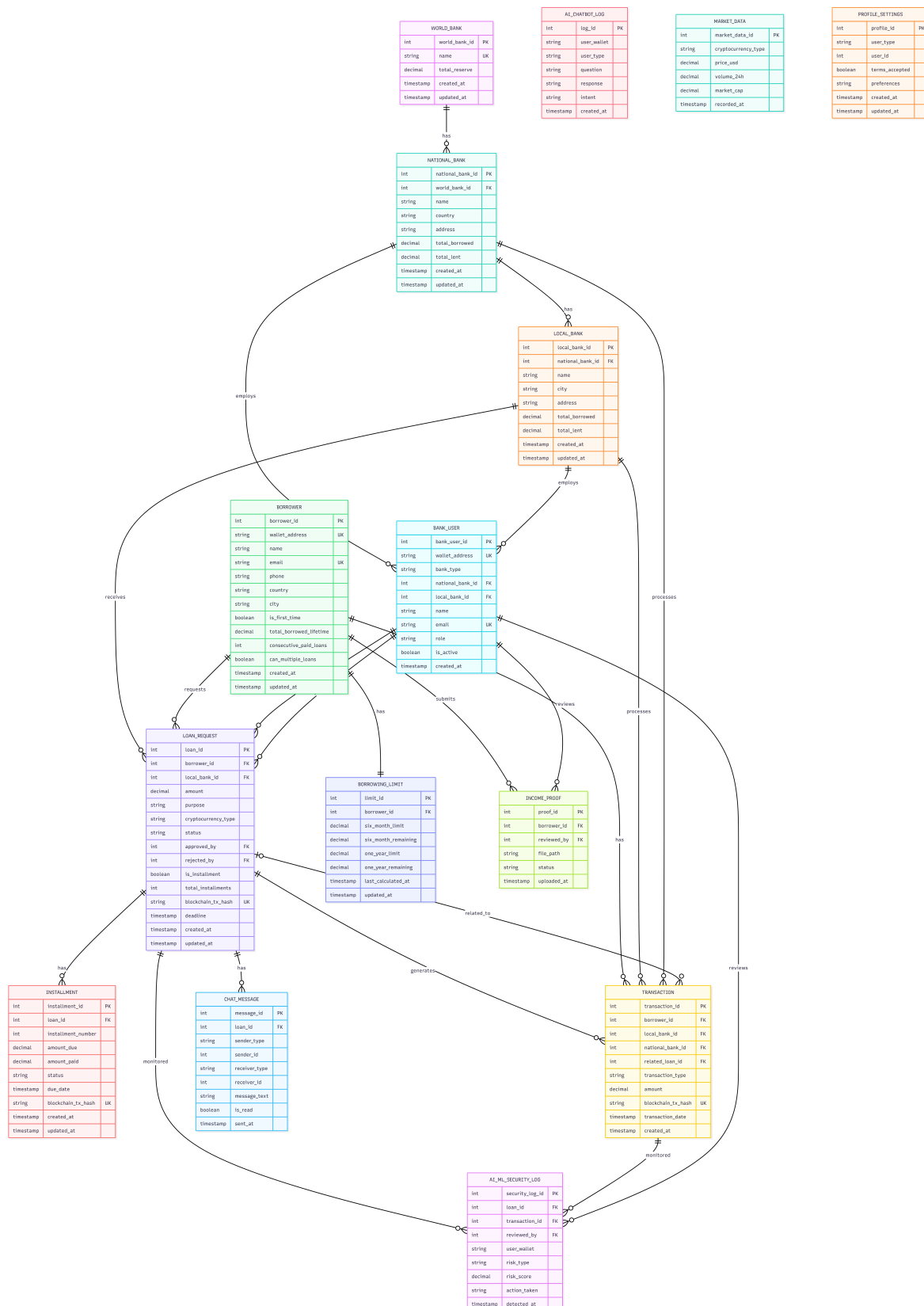


Figure 3.3: Entity-Relationship Diagram (ERD) for the Crypto World Bank database with relational symbols and entity attributes across all 15 tables.

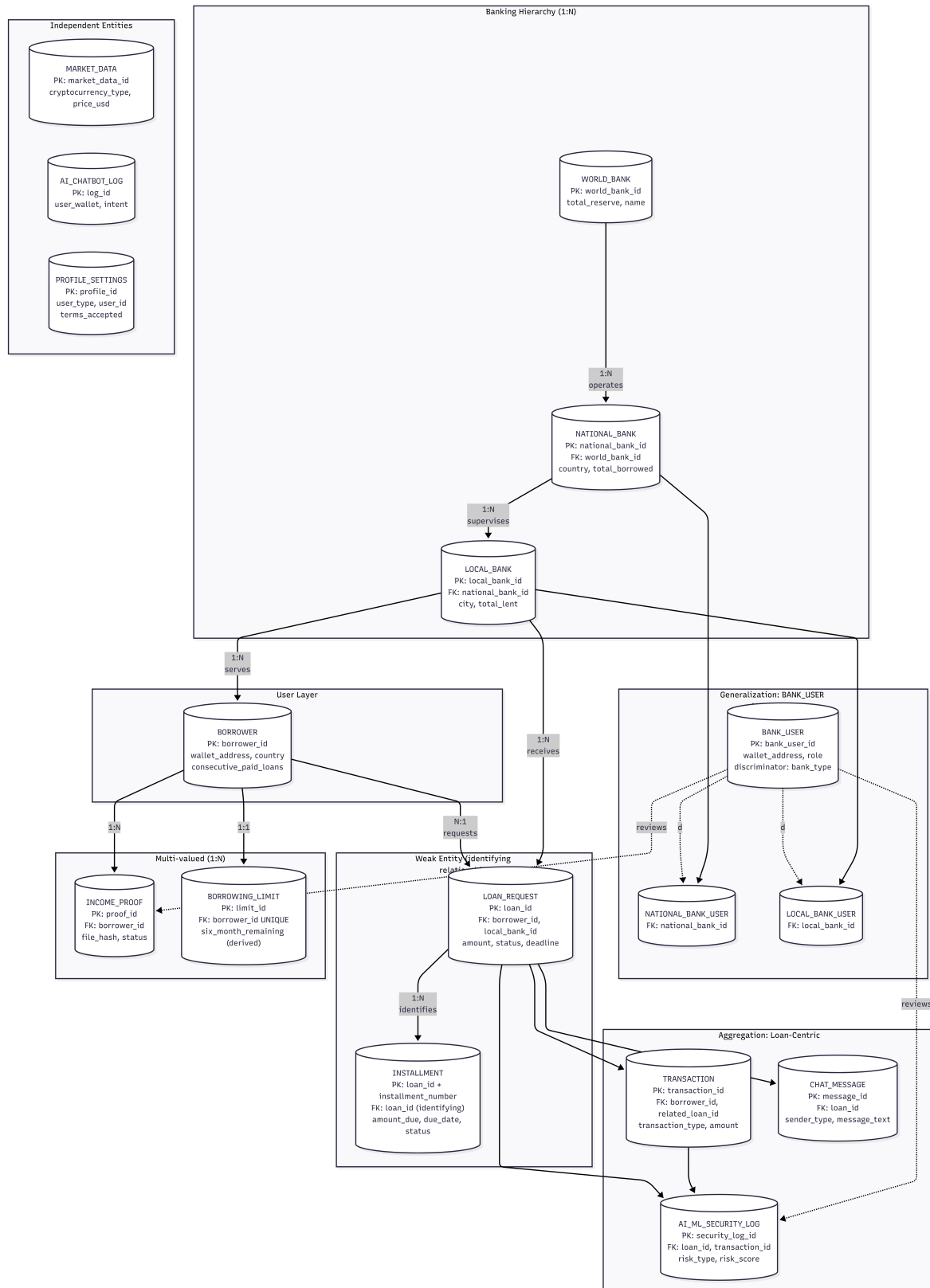


Figure 3.4: Enhanced Entity-Relationship (EER) diagram showing generalization/specialization, weak entities, aggregation, participation constraints, and cardinality for the complete data model.

3.3.1 Entity Summary

Table 3.2: Database entity summary (15 entities).

Entity	Role
WORLD_BANK	Top-level reserve holder; global lending parameters
NATIONAL_BANK	Country-level banks; borrow from World Bank
LOCAL_BANK	City-level banks; retail lending to borrowers
BANK_USER	Bank staff with role-based permissions (approve/reject loans)
BORROWER	End-users requesting and repaying loans
LOAN_REQUEST	Loan applications and full lifecycle tracking
INSTALLMENT	Installment payment records (weak entity dependent on LOAN_REQUEST)
TRANSACTION	Financial transaction records
BORROWING_LIMIT	Per-borrower limits with derived 6-month/1-year windows
INCOME_PROOF	Income verification documents (multi-valued per borrower)
CHAT_MESSAGE	Borrower–bank communication
AI_CHATBOT_LOG	AI chatbot interaction records
AI_ML_SECURITY_	Security and ML monitoring events
MARKET_DATA	Cryptocurrency price feeds
PROFILE_SETTING	User profile and platform preferences

3.3.2 EER Constructs Applied

Table 3.3: EER constructs applied in the Crypto World Bank data model.

EER Construct	Application
Generalization (disjoint)	BANK_USER generalizes to NATIONAL_BANK_USER and LOCAL_BANK_USER via <code>bank_type</code> discriminator with mutually exclusive foreign keys
Weak entity	INSTALLMENT is existence-dependent on LOAN_REQUEST with partial key <code>installment_number</code>
Aggregation	LOAN_REQUEST aggregates BORROWER + LOCAL_BANK; TRANSACTION, CHAT_MESSAGE, and AI_ML_SECURITY_LOG are components
Multi-valued attribute	INCOME_PROOF modeled as a separate entity (borrower can have multiple proofs)
Derived attribute	<code>six_month_remaining</code> and <code>one_year_remaining</code> in BORROWING_LIMIT are computed from TRANSACTION history
Composite key	INSTALLMENT uses (<code>loan_id</code> , <code>installment_number</code>)
Participation constraints	BORROWER has total participation in LOAN_REQUEST; INSTALLMENT has total participation when <code>is_installment = TRUE</code>
Cardinality	1:1 (BORROWER-BORROWING_LIMIT), 1:N (hierarchy, loan-installment), N:1 (loan-borrower)

3.3.3 Normalization

The schema is normalized to Third Normal Form (3NF) with verification against Boyce-Codd Normal Form (BCNF):

- **1NF:** All attributes are atomic with no repeating groups. Multi-valued attributes like income proofs are stored in separate entities.
- **2NF:** No partial dependencies exist. INSTALLMENT's non-key attributes depend on the full composite key (`loan_id`, `installment_number`).

- **3NF:** No transitive dependencies. Derived values such as `total_borrowed` in bank entities are computed at query time rather than stored redundantly.
- **BCNF:** All determinants are candidate keys. A CHECK constraint on `BANK_USER` enforces disjoint specialization of the generalization hierarchy.

3.3.4 Indexing Strategy

We use B-tree indexes (the default in PostgreSQL) for efficient retrieval on frequently queried columns:

Table 3.4: Database indexing strategy.

Table	Index	Purpose
LOAN_REQUEST	<code>idx_status_borrower</code> , <code>idx_status_bank</code> , <code>idx_deadline</code>	Filter pending loans; lookup by borrower or bank
TRANSACTION	<code>idx_type_date</code> , <code>idx_borrower_date</code> , <code>idx_date</code>	Borrowing limit queries (6-month/1-year windows)
BORROWER	<code>idx_wallet</code> , <code>idx_location</code>	Wallet address lookup; location-based queries
CHAT_MESSAGE	<code>idx_loan_sent</code> , <code>idx_sender</code> , <code>idx_receiver</code>	Chat history retrieval; unread message counts

B-tree indexes are chosen for their efficiency with range queries (e.g., transactions within the last 6 months); hash indexes would be suitable only for exact-match lookups.

3.3.5 Functional Dependencies

Table 3.5: Representative functional dependencies.

Relation	Functional Dependency	Notes
LOAN_REQUEST	<code>loan_id</code> $\rightarrow$ all attributes	Primary key determines row
LOAN_REQUEST	<code>borrower_id</code> , <code>local_bank_id</code> $\rightarrow$ <code>status</code> , <code>amount</code> , ...	One active request per borrower per bank
BORROWING_LIMIT	<code>borrower_id</code> $\rightarrow$ <code>six_month_limit</code> , <code>one_year_limit</code> , ...	1:1 with BORROWER

3.3.6 Relational Integrity Constraints

Table 3.6: Relational integrity constraints.

Constraint Type	Examples
Primary Key	<code>world_bank_id</code> , <code>loan_id</code> , <code>borrower_id</code> , etc.
Foreign Key	<code>local_bank_id</code> in LOAN_REQUEST references LOCAL_BANK
UNIQUE	<code>wallet_address</code> in BORROWER; <code>blockchain_tx_hash</code> in LOAN_REQUEST
CHECK	BANK_USER: (<code>bank_type</code> ='national' AND <code>national_bank_id</code> IS NOT NULL) OR (<code>bank_type</code> ='local' AND <code>local_bank_id</code> IS NOT NULL)
NOT NULL	Core attributes: <code>name</code> , <code>wallet_address</code> , <code>status</code>

3.4 On-Chain and Off-Chain Data Partitioning

Table 3.7: Data partitioning between on-chain and off-chain storage.

Data Category	Storage	Rationale
Reserve balances, loan requests, approval/rejection events, repayment transactions	On-chain	Immutability, public auditability, trustless verification
User profiles, income verification documents, chat messages, AI/ML inference logs	Off-chain (PostgreSQL)	Data privacy, query flexibility, storage cost optimization
Borrowing limit computations	Off-chain with on-chain enforcement	Complex temporal aggregation; results committed as on-chain constraints
Cryptocurrency market data	Off-chain (cached)	High-frequency updates; external API dependency

3.5 Digital Identity System

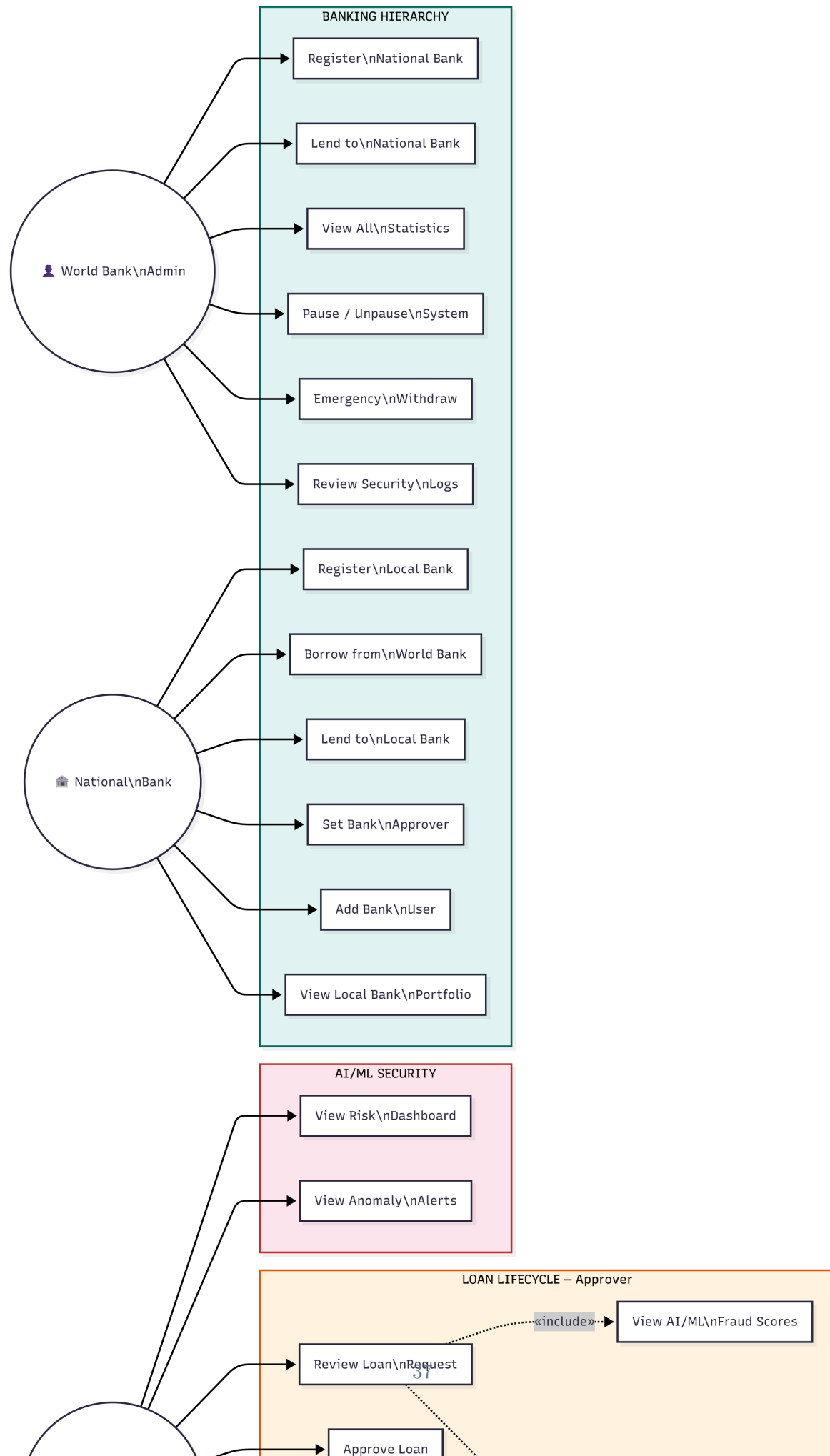
- **Wallet-based identity:** Users authenticate via Ethereum wallet signatures (MetaMask, WalletConnect). No centralized credential store is required.
- **Role binding:** Wallet addresses are mapped to hierarchical roles (Owner, National Bank, Local Bank, Approver, Borrower) within the smart contract permission system.

3.6 System Modeling

This section presents the system analysis diagrams that model the platform's structure, behavior, and data flow.

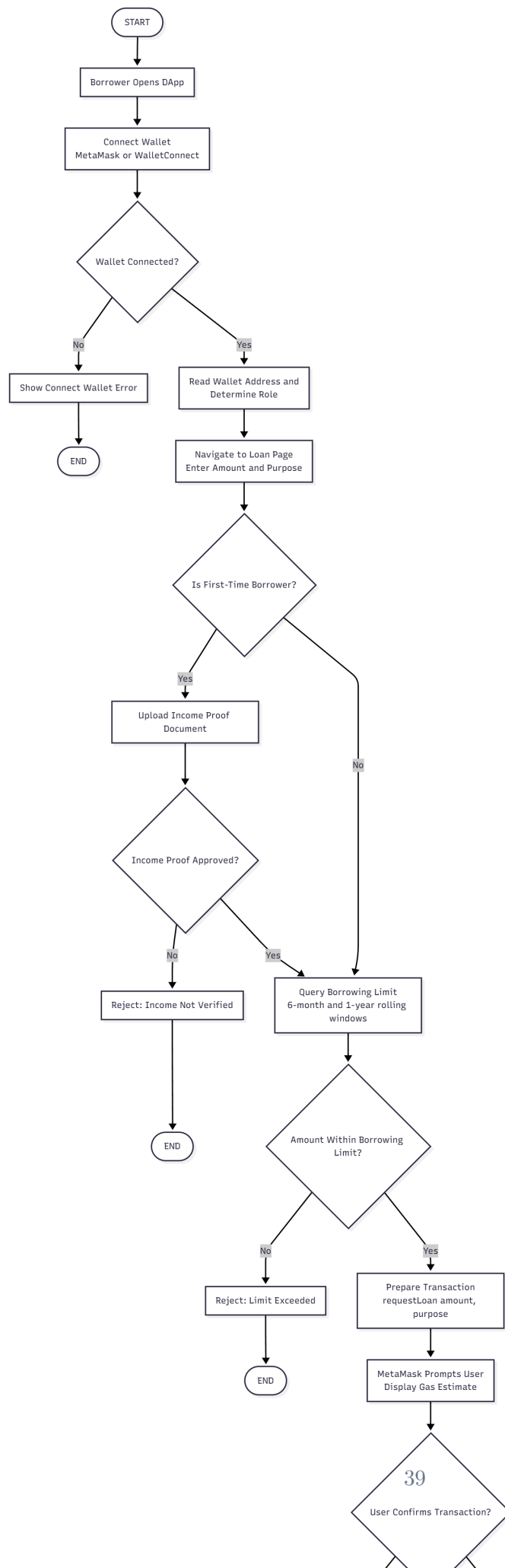
3.6.1 Use Case Diagram

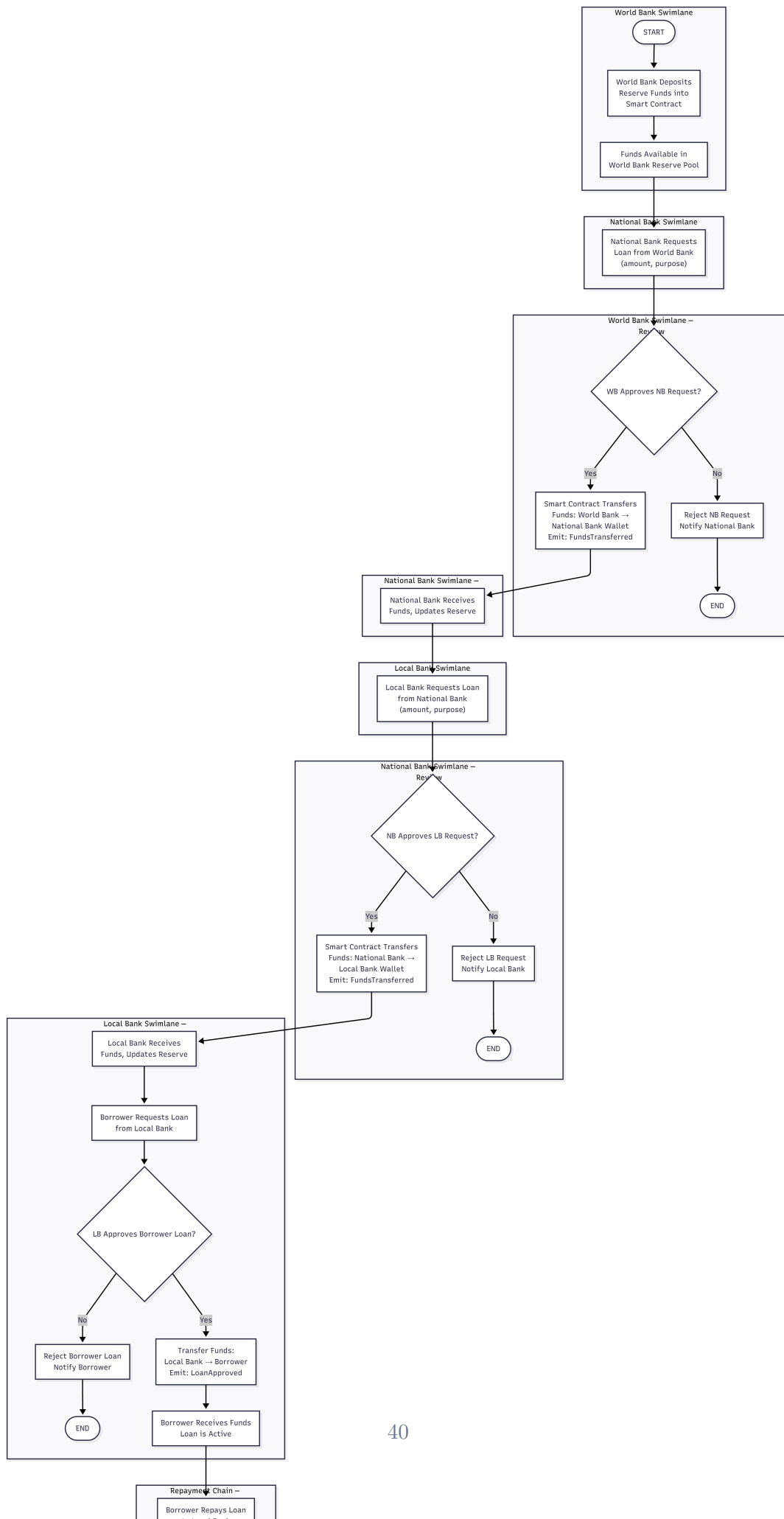
The system involves four primary actors—Borrower, Bank Approver, World Bank Admin, and National Bank—across 29 use cases. Figure 3.5 shows the complete use case diagram.

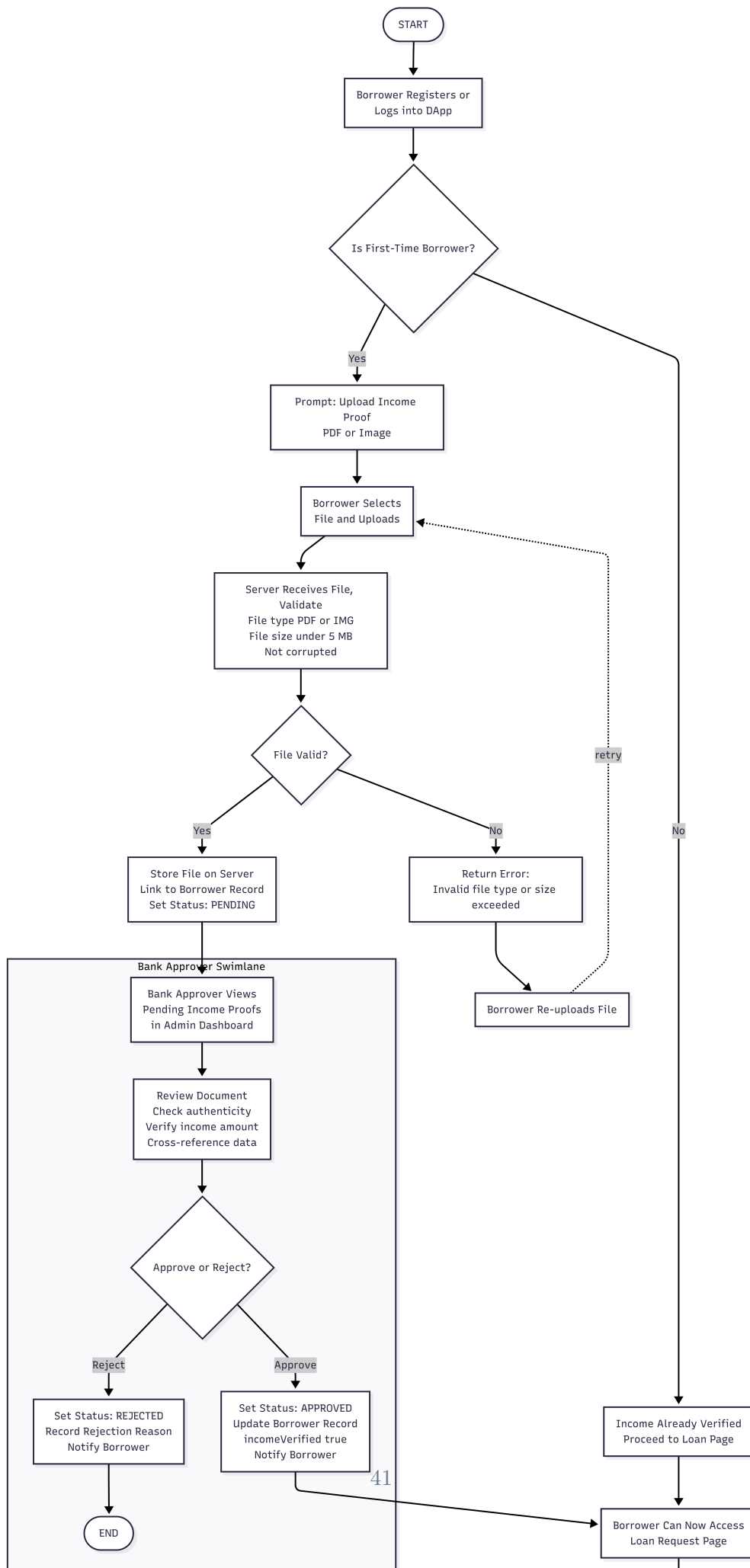


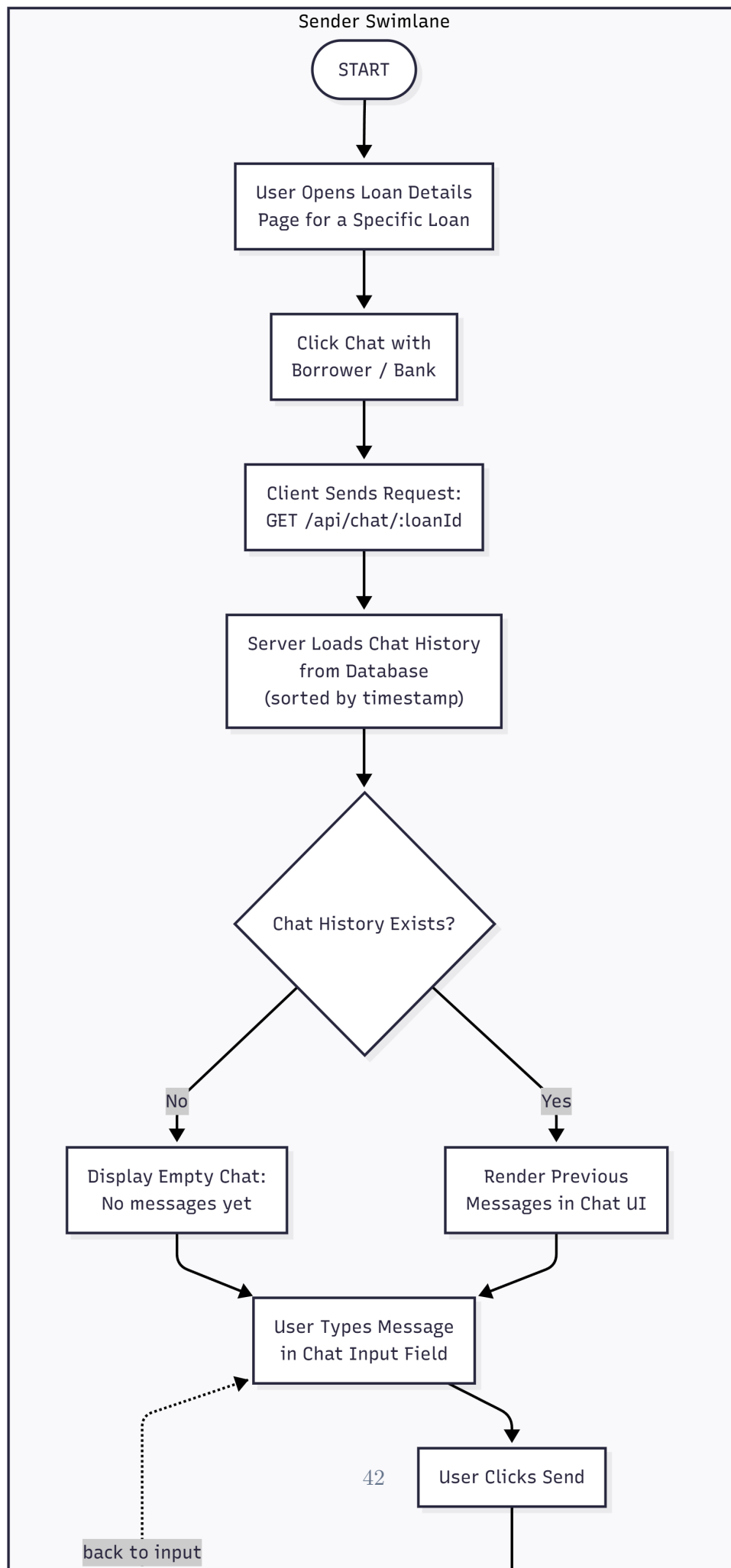
3.6.2 Activity Diagrams

Figures 3.6–3.12 present the activity diagrams for the platform’s seven core operational flows.









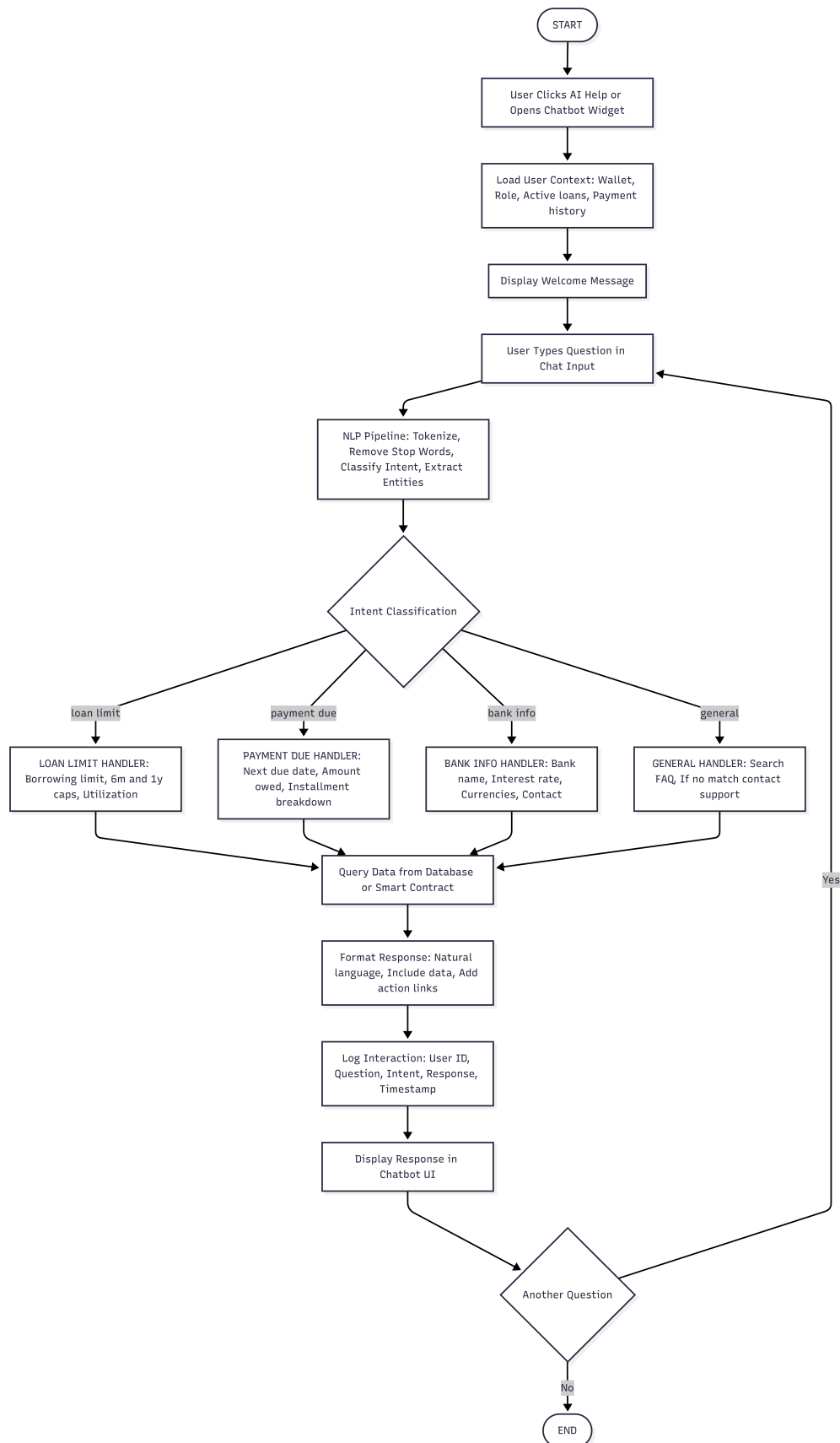
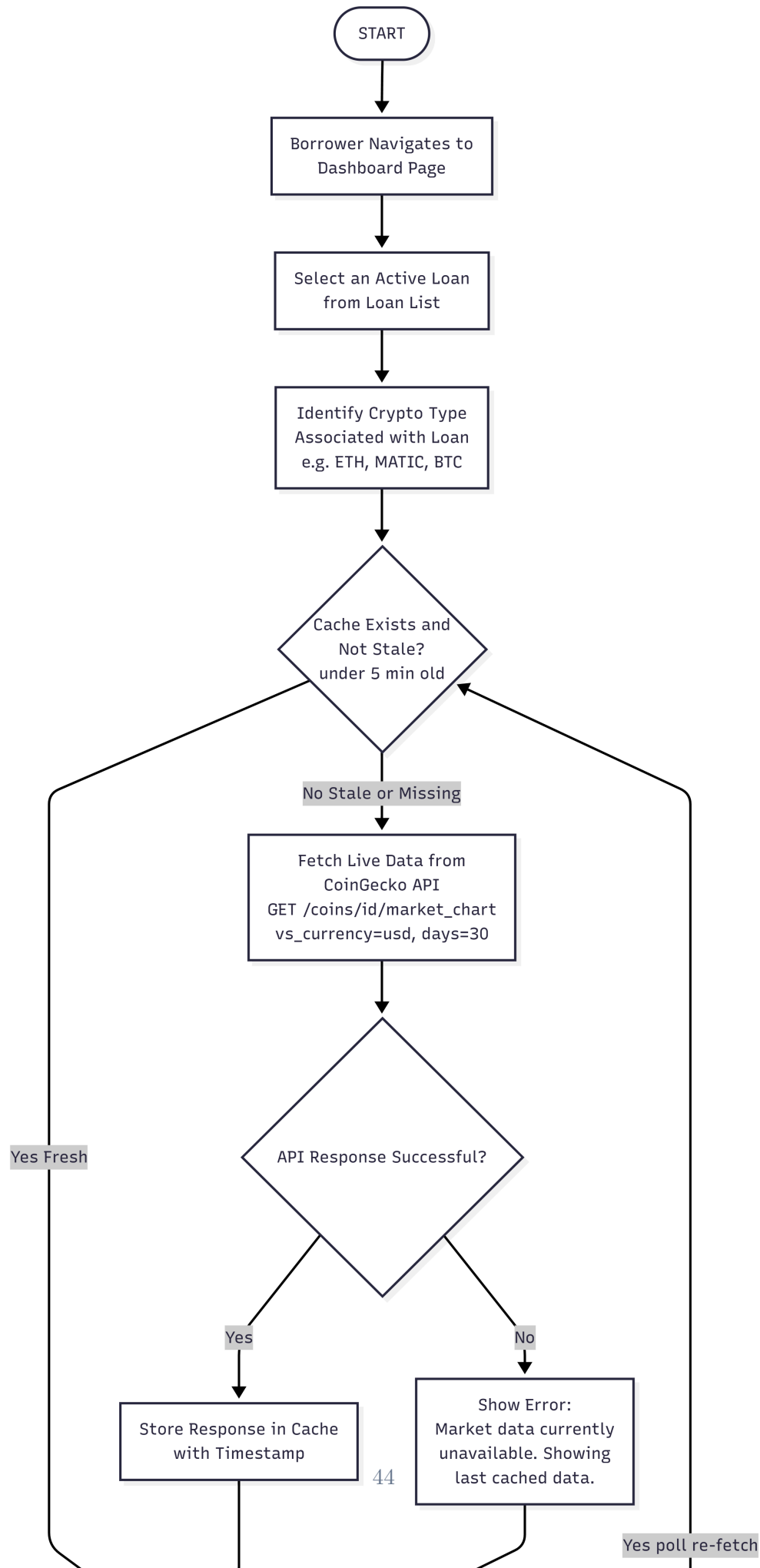


Figure 3.10: Activity diagram: AI chatbot interaction flow for automated borrower assistance and FAQ handling.



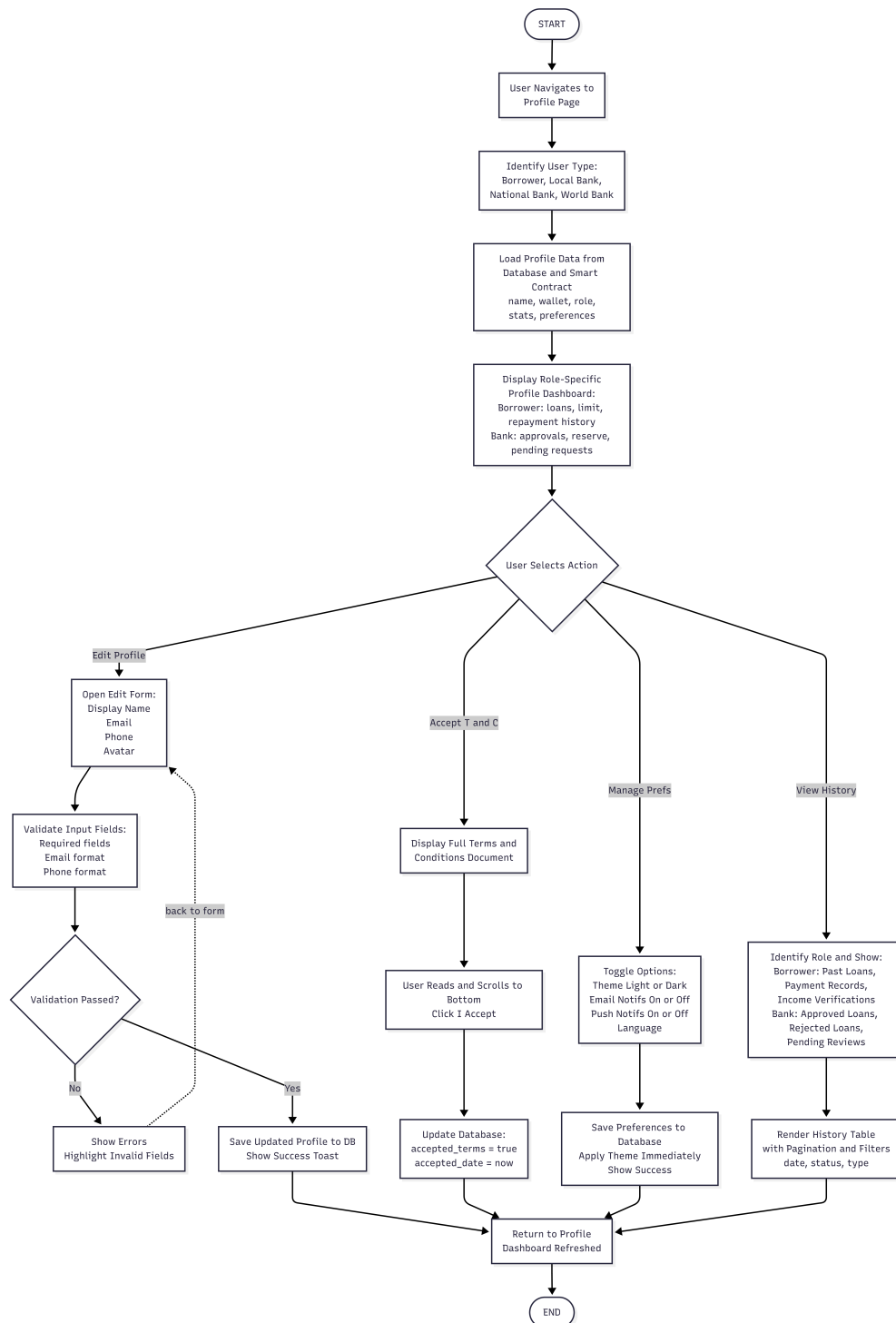


Figure 3.12: Activity diagram: Profile management flow for user settings, terms acceptance, and account configuration.

3.6.3 Data Flow Diagrams

Figures 3.13–3.15 present the data flow diagrams at the context level (Level 0) and decomposed level (Level 1).

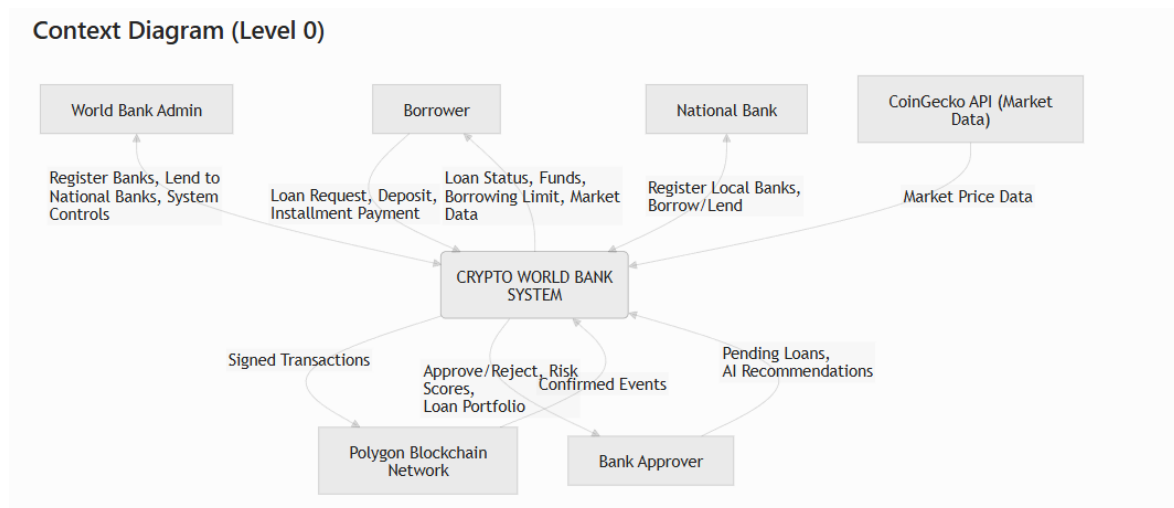


Figure 3.13: Data flow diagram — Context diagram (Level 0) showing external entities and data flows into and out of the Crypto World Bank system.

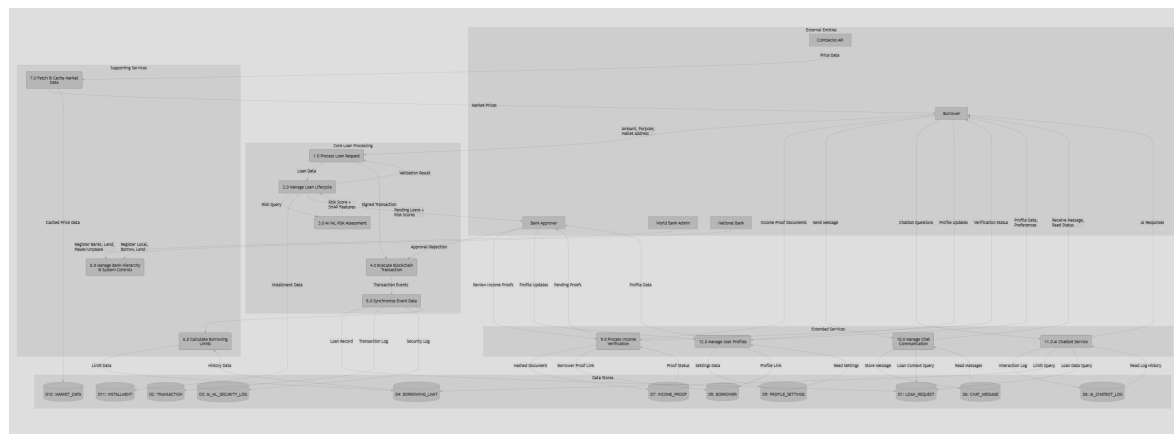


Figure 3.14: Data flow diagram — Level 1 (Part 1) showing decomposed processes, data stores, and information movement across system components.

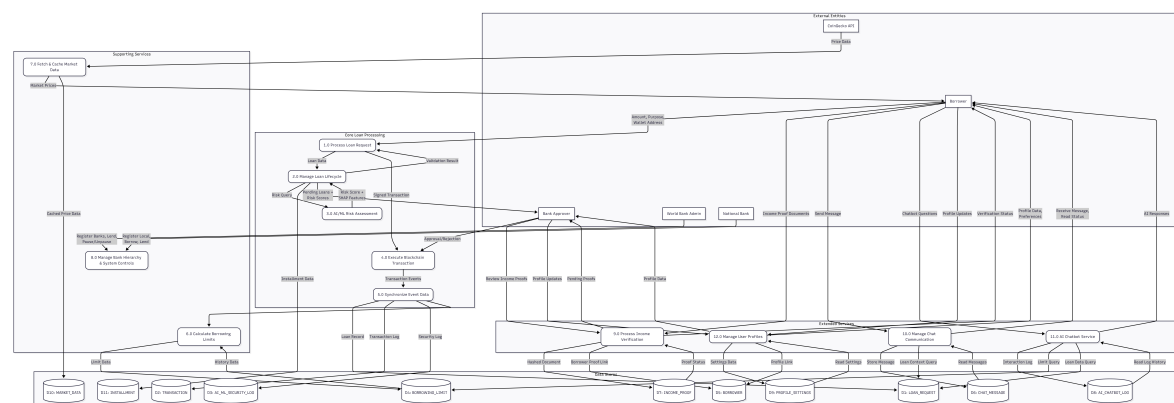


Figure 3.15: Data flow diagram — Level 1 (Part 2) showing additional processes for AI/ML analytics, market data, and profile management.

3.6.4 Sequence Diagrams

Figures 3.16–3.24 present the sequence diagrams for the platform’s key interaction flows.

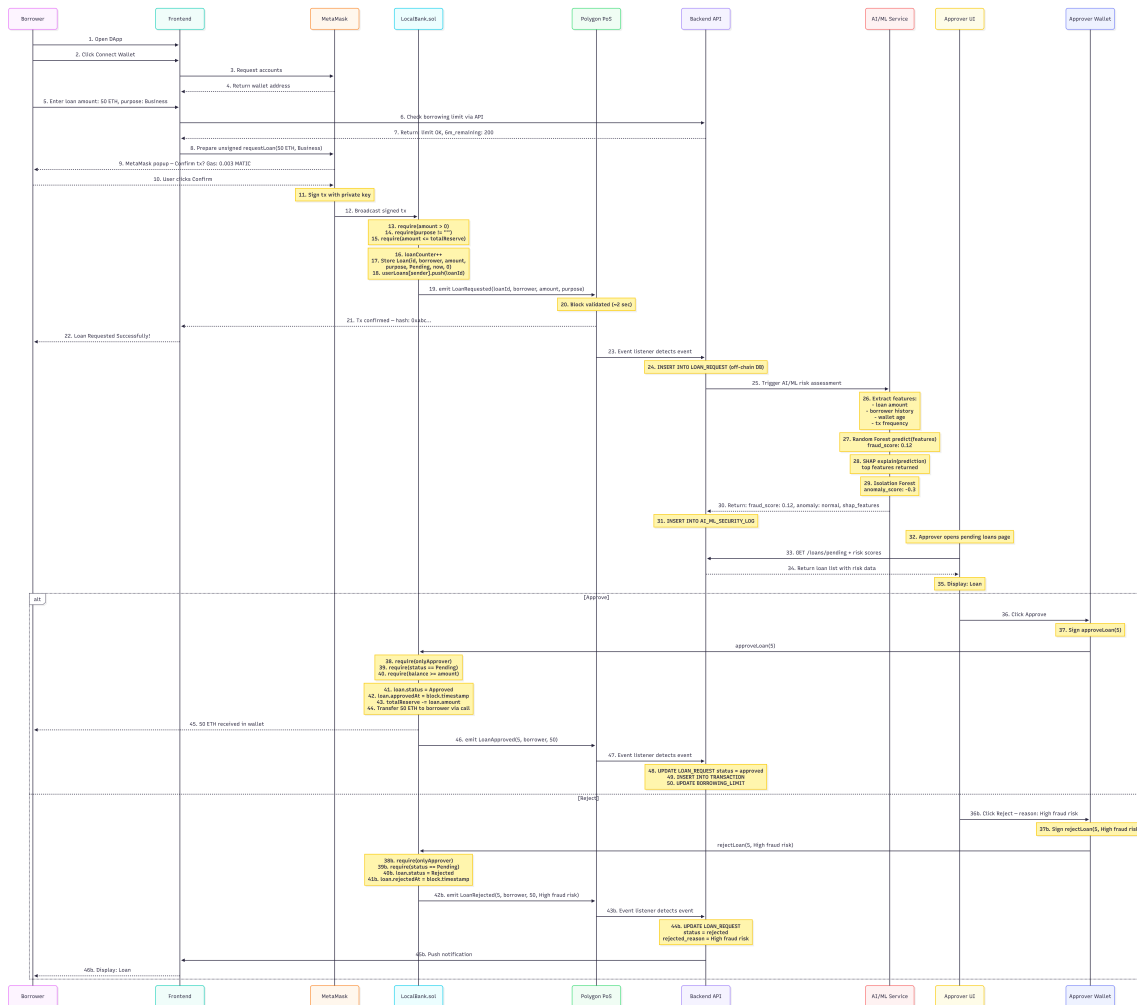


Figure 3.16: Sequence diagram: Loan request, AI risk check, and approval decision across 9 participants.

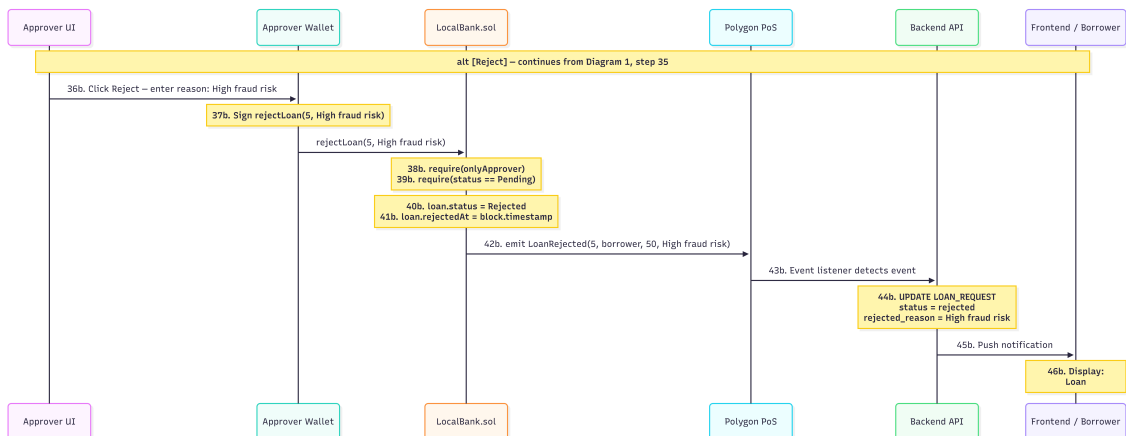


Figure 3.17: Sequence diagram: Reject path (alternative flow) for loan request denial.

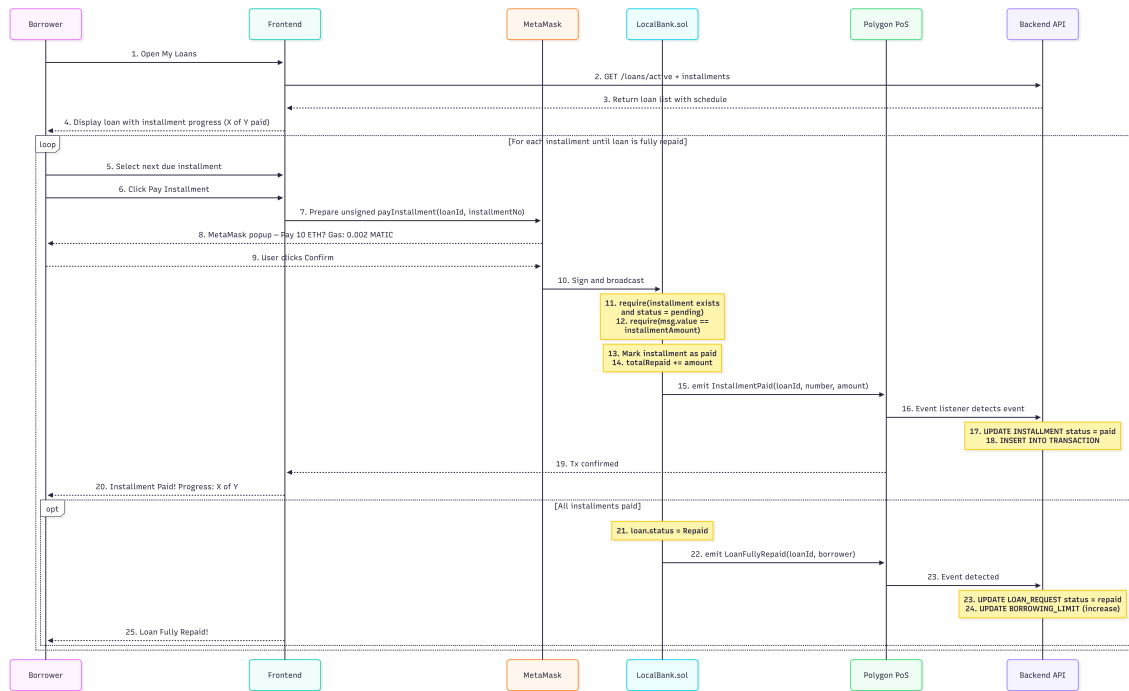


Figure 3.18: Sequence diagram: Installment payment loop showing recurring payment processing and balance updates.

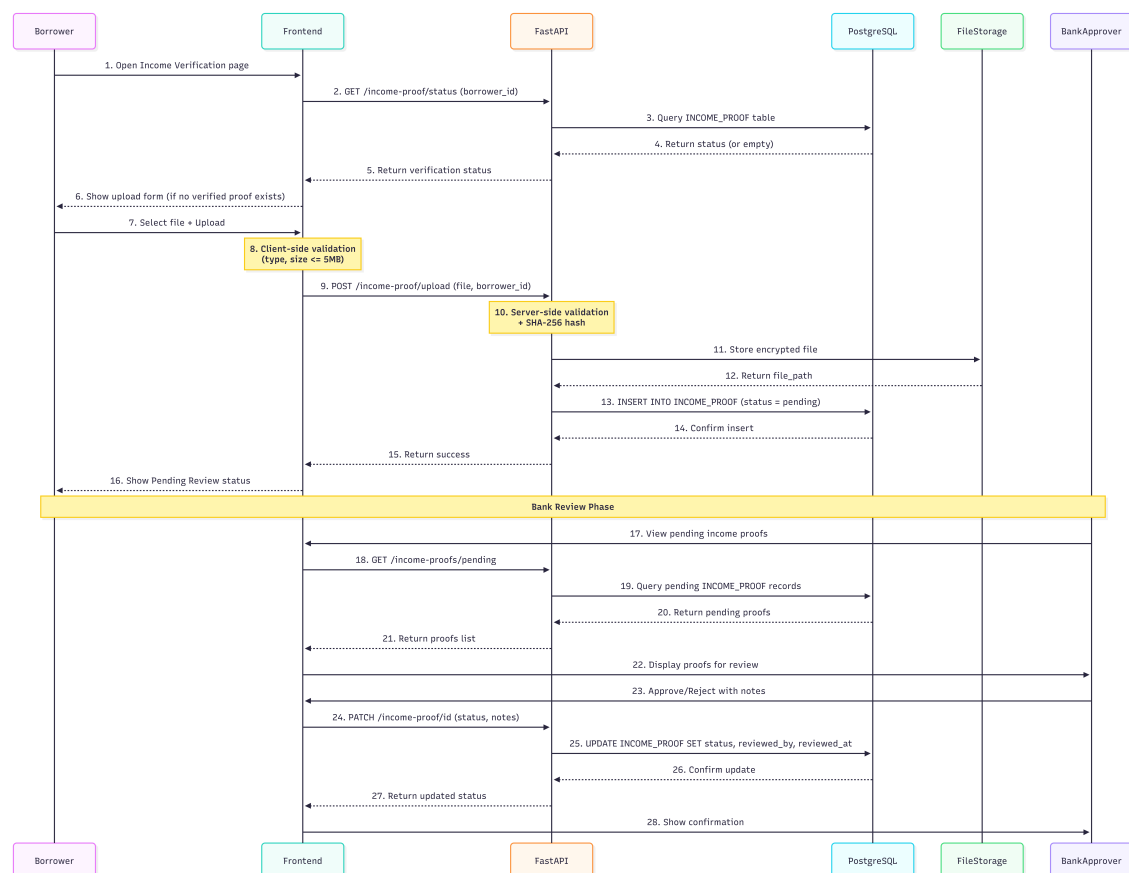


Figure 3.19: Sequence diagram: Income verification process between borrower, frontend, backend API, and database.

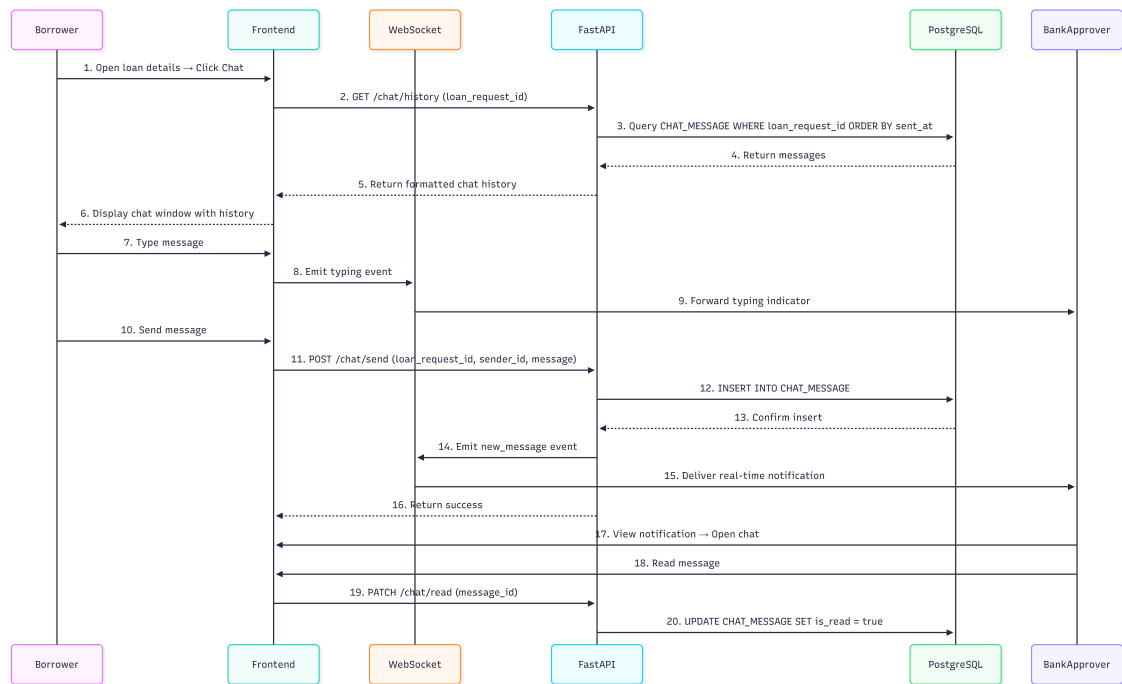


Figure 3.20: Sequence diagram: Chat system message exchange between borrower and bank representative.

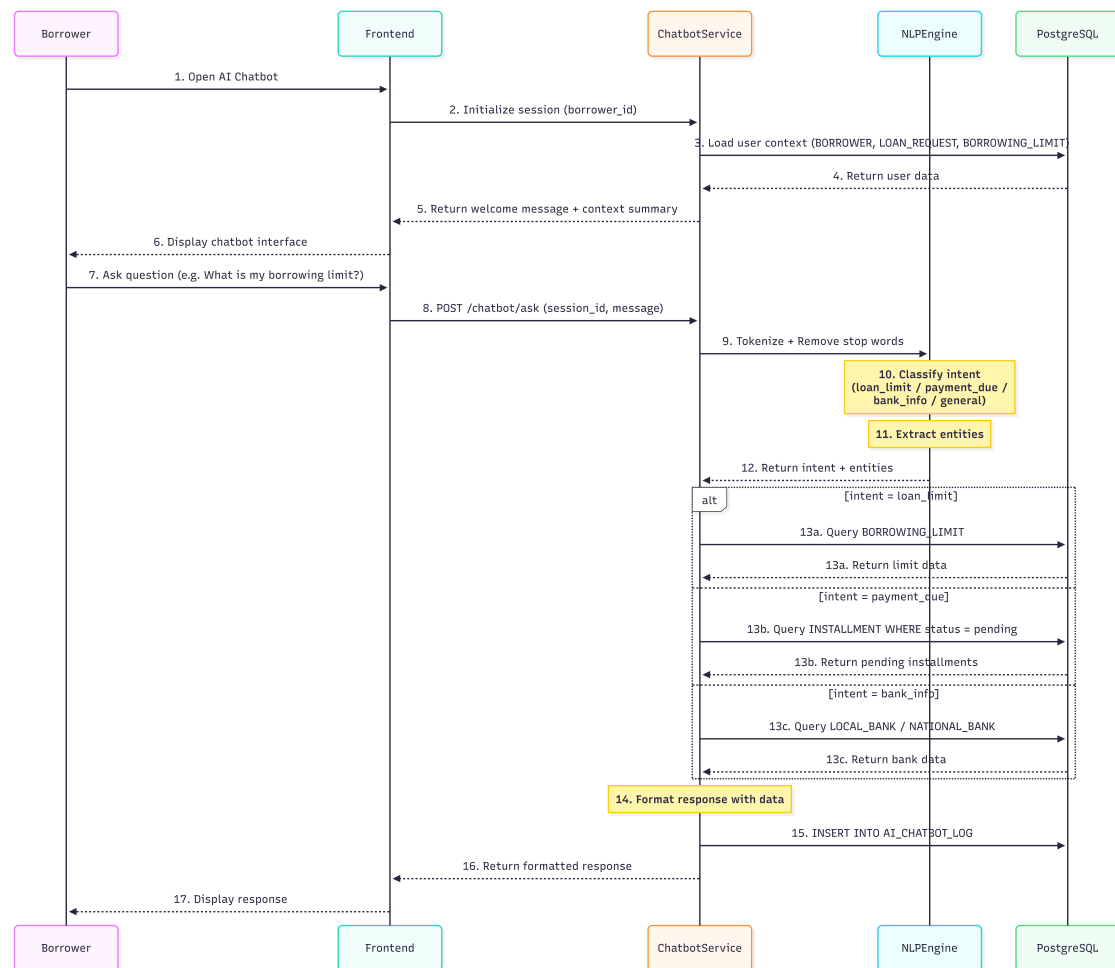


Figure 3.21: Sequence diagram: AI chatbot interaction for automated borrower support queries.

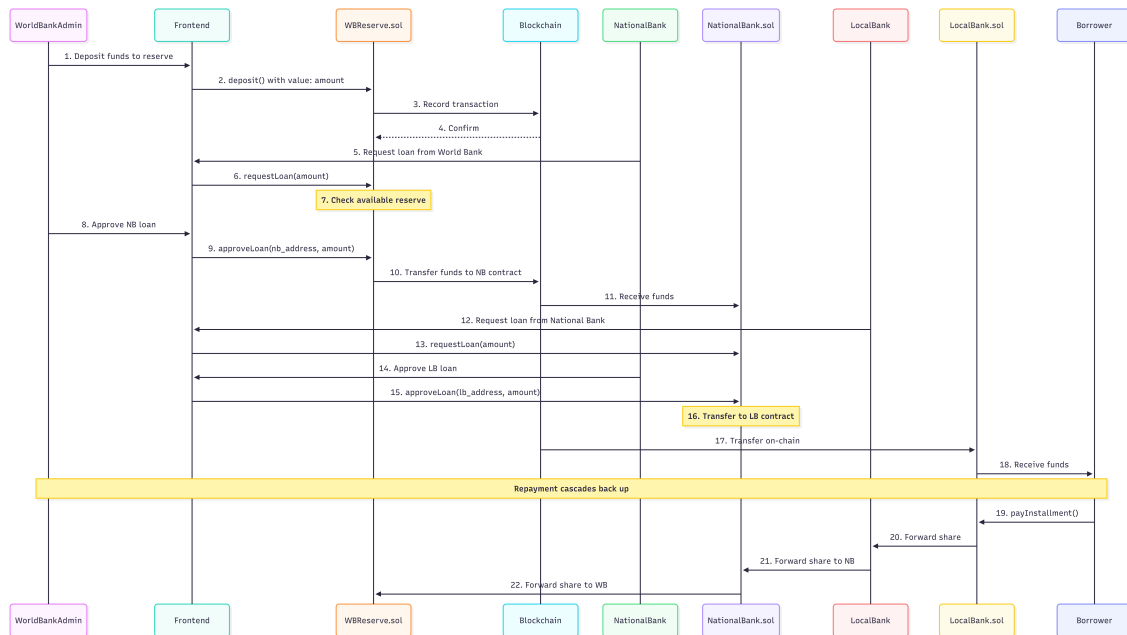


Figure 3.22: Sequence diagram: Hierarchical banking flow showing capital distribution from World Bank through National and Local Banks to Borrowers.

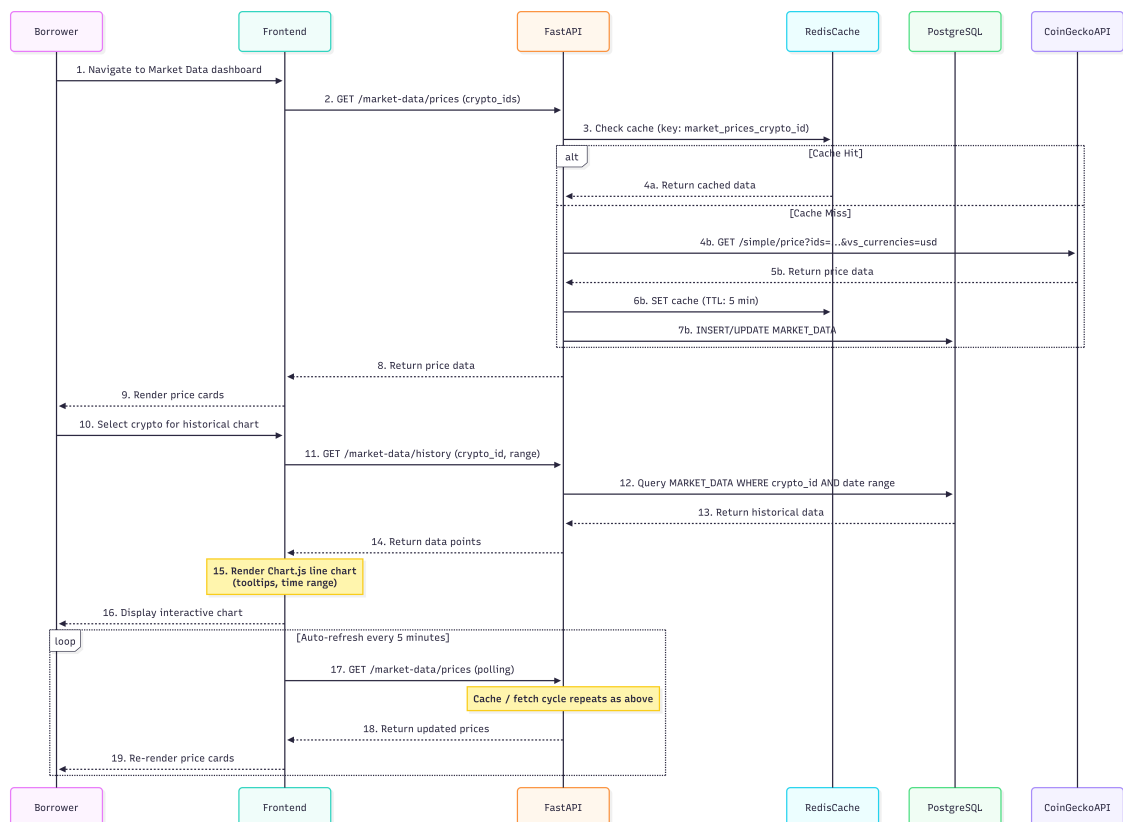


Figure 3.23: Sequence diagram: Market data retrieval from external cryptocurrency price APIs.

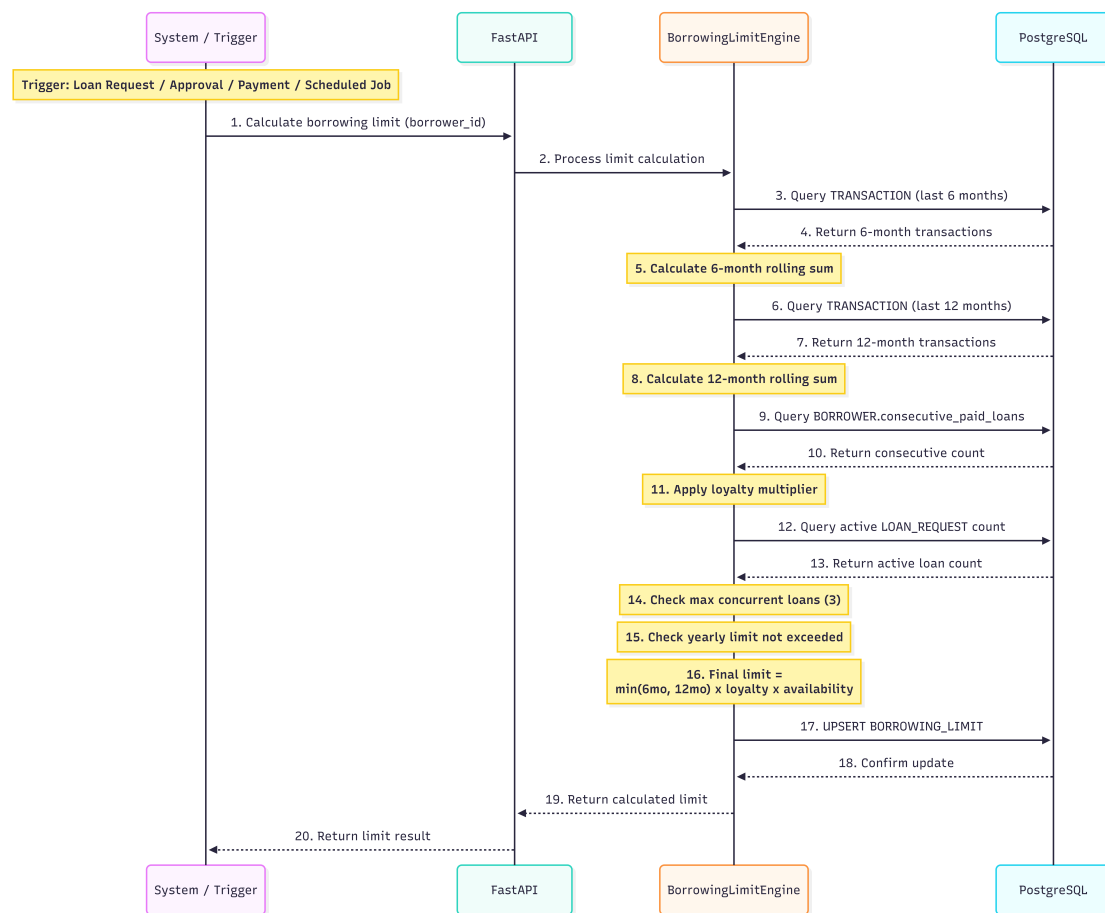


Figure 3.24: Sequence diagram: Borrowing limit calculation using rolling 6-month and 1-year transaction windows.

3.6.5 Four-Tier Capital Flow

Figure 3.25 illustrates the hierarchical capital flow and cascading repayment structure.

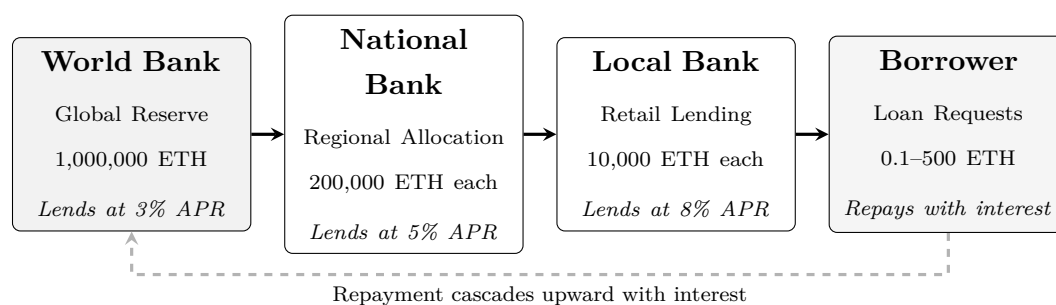


Figure 3.25: Four-tier hierarchical capital flow with cascading repayment.

3.7 Auxiliary Dual-Currency Facility

The Crypto World Bank's cryptocurrency exchange capability is designed as an **auxiliary facility** integrated into existing banking infrastructure worldwide. Rather than operating as a standalone exchange, the platform extends the services of participating

banks by enabling a dual-currency feature—allowing eligible bank customers to transact in both fiat and cryptocurrency within their existing banking relationship.

- **Integration model:** The dual-currency facility is offered by all participating banks as an add-on service to their existing product portfolio. No new banking license or standalone entity is required.
- **Eligibility determination:** Eligibility for dual-currency services is determined by the bank officers who manage lending operations, based on existing KYC/AML compliance, account standing, and lending relationship history.
- **No defaulting of conditions:** The project does not override, modify, or default any existing banking conditions, regulatory requirements, or contractual obligations.
- **Scope:** The facility enables fiat-to-crypto and crypto-to-fiat conversions, cryptocurrency-denominated lending and repayment, and transparent on-chain transaction records—all within the governance structure of the participating bank.

3.8 Governance Framework

3.8.1 Network Membership Governance

Table 3.8: Network membership governance.

Governance Aspect	Implementation
Member on-boarding	World Bank owner registers National Banks; National Banks register Local Banks; Local Banks designate approvers — all enforced on-chain
Member off-boarding	Deactivation flags in smart contracts; cascading access revocation
Regulatory oversight	Audit log emission via smart contract events; planned read-only regulator dashboard
Permission structure	Hierarchical: Owner → National Bank → Local Bank → Approver → Borrower; enforced by on-chain role-check modifiers
Network operations	Pause/unpause mechanism for emergency response; emergency withdrawal for critical situations

3.8.2 Business Network Governance

Table 3.9: Business network governance.

Governance Aspect	Implementation
Business charter	Defined in project documentation; operational parameters codified in smart contract constants
Common services	Reserve management, loan lifecycle orchestration, event-driven notification system
Business SLA	Testnet phase: best-effort availability. Production phase: 99.5% target uptime with multi-region deployment
Regulatory compliance	Architecture designed for audit trail generation; data partitioning supports GDPR-style data subject requests

3.8.3 Technology Infrastructure Governance

Table 3.10: Technology infrastructure governance.

Governance Aspect	Implementation
Distributed IT structure	Client-side frontend (decentralized delivery); blockchain layer (fully decentralized); backend API (centralized, horizontally scalable)
Technology assessment	Continuous evaluation of EVM alternatives (L2 rollups, sidechains) for cost and performance optimization
On-chain / off-chain data services	Clearly partitioned (see Table 3.5); event listeners synchronize state between layers
Risk mitigation	Smart contract pause mechanism; ReentrancyGuard; input validation; planned formal security audit

3.8.4 Regulatory Compliance Considerations

As the platform operates within the regulated banking domain:

- Our prototype phase operates exclusively on **public testnets** with no real monetary value.
- The architecture supports **audit log generation** for regulatory review.
- Future production deployment would engage **regulatory sandbox programs** in target jurisdictions.

3.8.5 Asset Tokenization

- **Current implementation:** Native blockchain currency (ETH/MATIC) serves as the reserve and loan denomination.
- **Planned extension:** ERC-20 stablecoin support (USDC, USDT) for USD-denominated lending operations; tokenized collateral instruments for under-collateralized lending scenarios.

Chapter 4

Methodology

This project and its planning have been structured in accordance with the [Blockchain Olympiad Bangladesh AI guidelines](#) [16] and the final-year project evaluation rubrics.

4.1 Development Methodology

We adopted a **lightweight Agile/Scrum** methodology tailored for an academic prototype with a fixed two-month development window. The iterative approach enables incremental delivery of demonstrable features while accommodating evolving requirements inherent in research-oriented development. Figure 4.1 illustrates our Agile process.

5.6 Agile/Scrum Process Flow

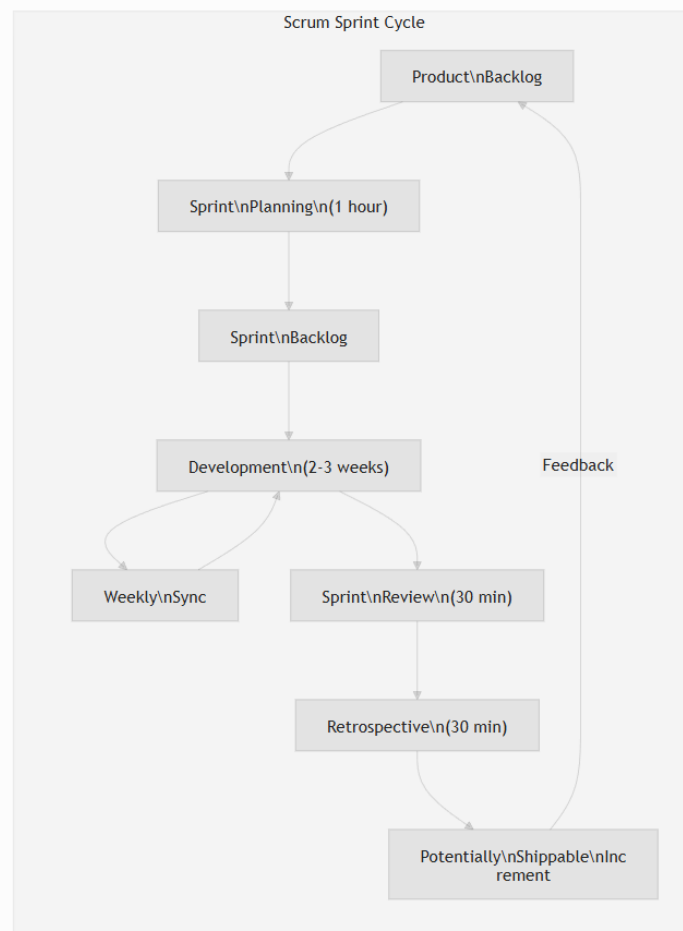


Figure 4.1: Agile/Scrum process diagram showing sprint cycles, backlog refinement, and iterative delivery within the 8-week development window.

Key parameters of our Agile implementation:

- **Sprint Duration:** 2–3 weeks per sprint (3 sprints total)
- **Team Size:** 2 developers (thesis project)
- **Weekly Sync:** Progress review, blocker identification, and planning (replaces daily standup for a 2-person team)
- **Sprint Planning:** 1 hour at sprint start
- **Sprint Review and Retrospective:** 30 minutes each at sprint end

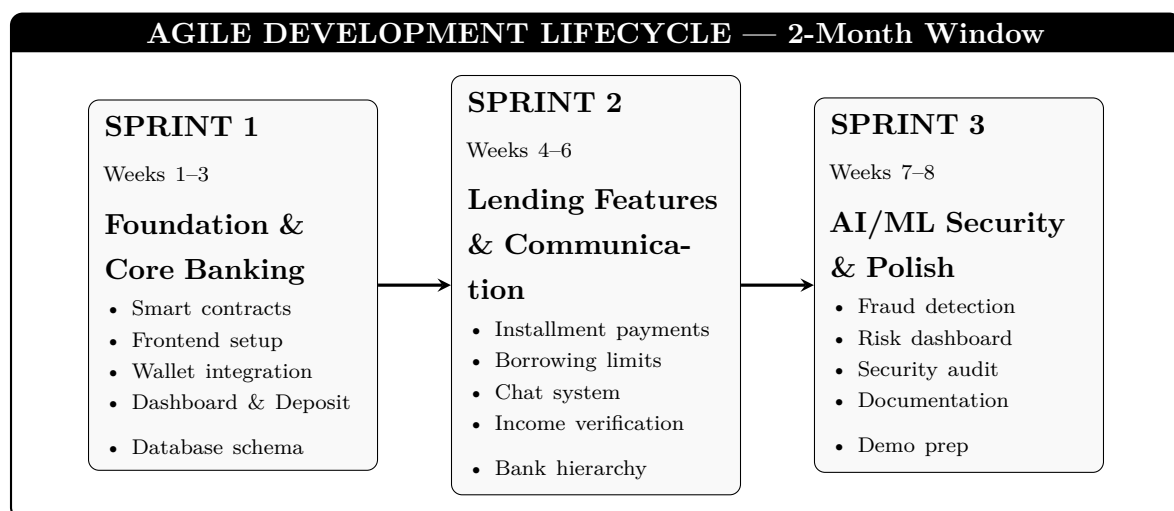
4.2 Planned AI/ML Support

For pre-thesis 1, the AI/ML component is treated as a supporting layer rather than the central contribution of the project. Its purpose is to show how fraud monitoring, anomaly detection, and explainable decision support can be integrated into the lending workflow without making the current paper depend on a completed data-science pipeline.

At this stage, the planned AI/ML scope is intentionally modest:

- **Fraud monitoring:** Random Forest is considered as a practical candidate model for transaction-risk screening.
- **Anomaly detection:** Isolation Forest is proposed for identifying unusual wallet behavior where labeled data may be limited.
- **Explainability:** SHAP-based feature attribution is retained so that flagged decisions remain interpretable to human reviewers.

Full dataset construction, extensive model benchmarking, and production-grade validation are reserved for later thesis stages and future work, since the primary contribution of this pre-thesis report is the hierarchical blockchain lending architecture rather than a completed machine-learning study.



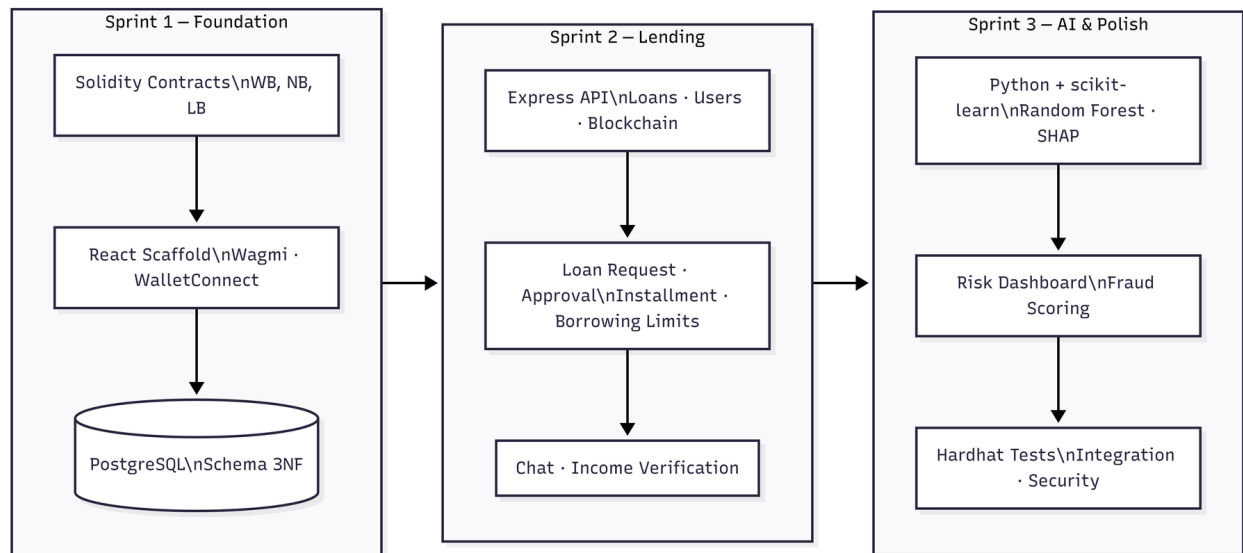


Figure 4.2: Technical development methodology: blockchain (Solidity), frontend (React, Wagmi, WalletConnect), backend (Express API, PostgreSQL), and AI/ML (Python, scikit-learn, Random Forest, SHAP) components across the three sprints.

4.3 Sprint Plan and Deliverables

4.3.1 Sprint 1: Foundation and Core Banking (Weeks 1–3)

Sprint Goal: Establish core blockchain infrastructure and hierarchical banking structure.

Table 4.1: Sprint 1 backlog: Smart contract development (21 pts).

ID	User Story	Pts
US-1.1	World Bank contract — reserve management, national bank registration	5
US-1.2	National Bank contract — borrow from WB, lend to Local Banks	5
US-1.3	Local Bank contract — borrow from NB, lend to users	5
US-1.4	Role-based access control — WB/NB/LB Admin, Bank User, Borrower	3
US-1.5	Gas cost management — initiator pays; Polygon low-fee (\$0.001–0.01/tx)	3

Table 4.2: Sprint 1 backlog: Frontend foundation (13 pts) and database schema (8 pts).

ID	User Story	Pts
US-1.6	Wallet connection — MetaMask, WalletConnect, Sepolia/Mumbai	3
US-1.7	Dashboard UI — Material Design 3, responsive, blockchain-themed	5
US-1.8	Navigation and layout — AppBar, role-based menu	3
US-1.9	Blockchain visual elements — tx hash display, security badges	2
US-1.10	Database design — 15 tables, 3NF	5
US-1.11	Migration scripts and seed data	3

Sprint 1 Total: 42 story points. Deliverables: smart contracts deployed on testnet, basic frontend with wallet connection, database schema implemented, role-based access control, dashboard UI.

4.3.2 Sprint 2: Lending Features and Communication (Weeks 4–6)

Sprint Goal: Implement complete lending workflow and communication features.

Table 4.3: Sprint 2 backlog: Lending and communication features (50 pts).

ID	User Story	Pts
US-2.1	Loan request submission with blockchain transaction	5
US-2.2	Loan approval and rejection workflow with bank approver	5
US-2.3	Installment payment system with automated schedule generation	8
US-2.4	Borrowing limit engine (6-month and 1-year rolling windows)	5
US-2.5	Borrower–bank real-time chat system	5
US-2.6	Income verification document upload and review	5
US-2.7	Hierarchical bank registration (National → Local)	5
US-2.8	Bank user management and approver designation	3
US-2.9	Loan history and transaction tracking pages	5
US-2.10	QR code generation for wallet addresses	2
US-2.11	Responsive UI polish and error handling	2

4.3.3 Sprint 3: AI/ML Security and Finalization (Weeks 7–8)

Sprint Goal: Integrate AI/ML analytics, complete testing, and prepare documentation.

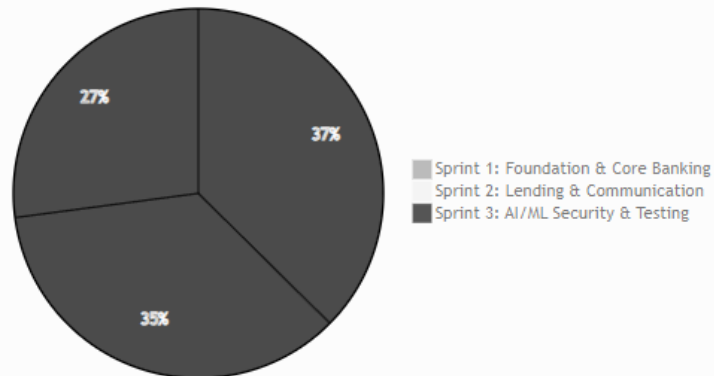
Table 4.4: Sprint 3 backlog: AI/ML security and finalization (38 pts).

ID	User Story	Pts
US-3.1	Random Forest fraud detection model training and deployment	8
US-3.2	SHAP-based explainability for risk assessments	5
US-3.3	Isolation Forest anomaly detection for wallet behavior	5
US-3.4	Risk dashboard with real-time AI/ML scores	5
US-3.5	AI chatbot for borrower assistance	3
US-3.6	Market data visualization page	3
US-3.7	Profile and settings management	2
US-3.8	Security audit and vulnerability assessment	3
US-3.9	Documentation and report finalization	2
US-3.10	Demo preparation and presentation	2

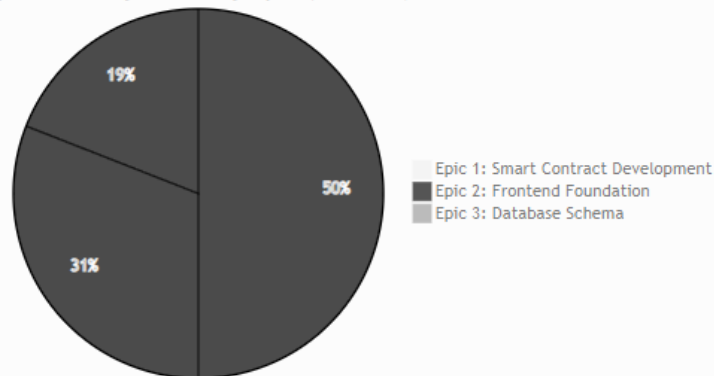
Figure 4.3 shows the sprint submission workflow.

5.5 Sprint Story Points Distribution

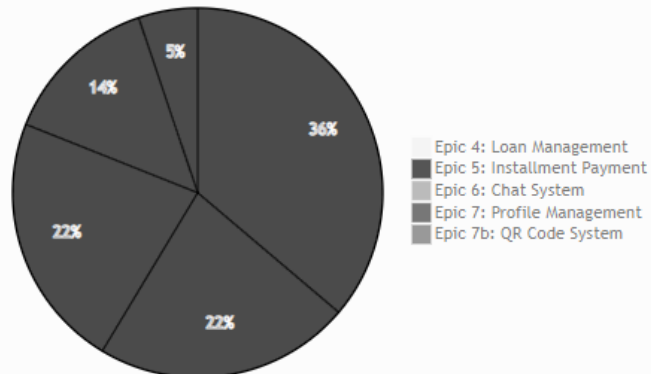
Points Distribution Across Sprints (155 Total)



Sprint 1 Story Points by Epic (42 Total)



Sprint 2 Story Points by Epic (58 Total)



Sprint 3 Story Points by Epic (55 Total)

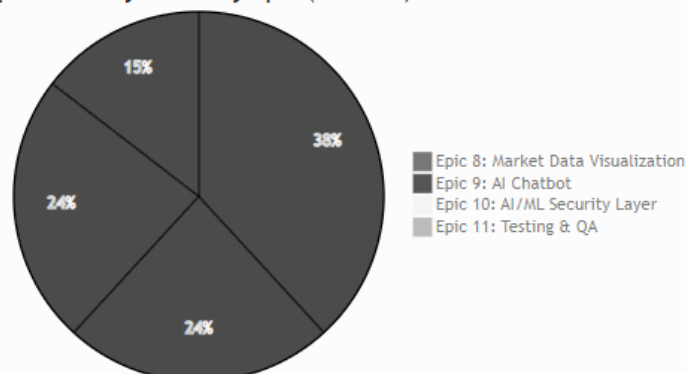


Figure 4.3: Sprint submission diagram showing the workflow from backlog refinement through development, testing, review, and delivery for each sprint cycle.

4.4 SDLC Stage Mapping

We map our Agile sprints to the seven stages of the Software Development Life Cycle (SDLC). Figure 4.4 illustrates this mapping.

Table 4.5: SDLC stage mapping.

SDLC Stage	Project Activity	Deliverable
1. Planning	Feasibility studies; professor consultations; guideline review	Feasibility Report; Project Plan
2. Requirements	System analysis; use case definitions; constraints identification	Use case diagrams; UC-1 through UC-5
3. Design	Three-layer architecture; DB schema (3NF); smart contract interfaces	Architecture diagrams; ERD; DFD
4. Development	Sprint 1–3: smart contracts, frontend, backend, AI/ML	Source code; DApp prototype
5. Testing	Hardhat unit tests (12+); integration testing; AI/ML evaluation	Test reports; model metrics
6. Deployment	Testnet deployment; frontend (Vercel); backend (Render)	Live prototype on testnet
7. Maintenance	Monitoring; model retraining; bug fixes; iteration	Updated docs; retrained models

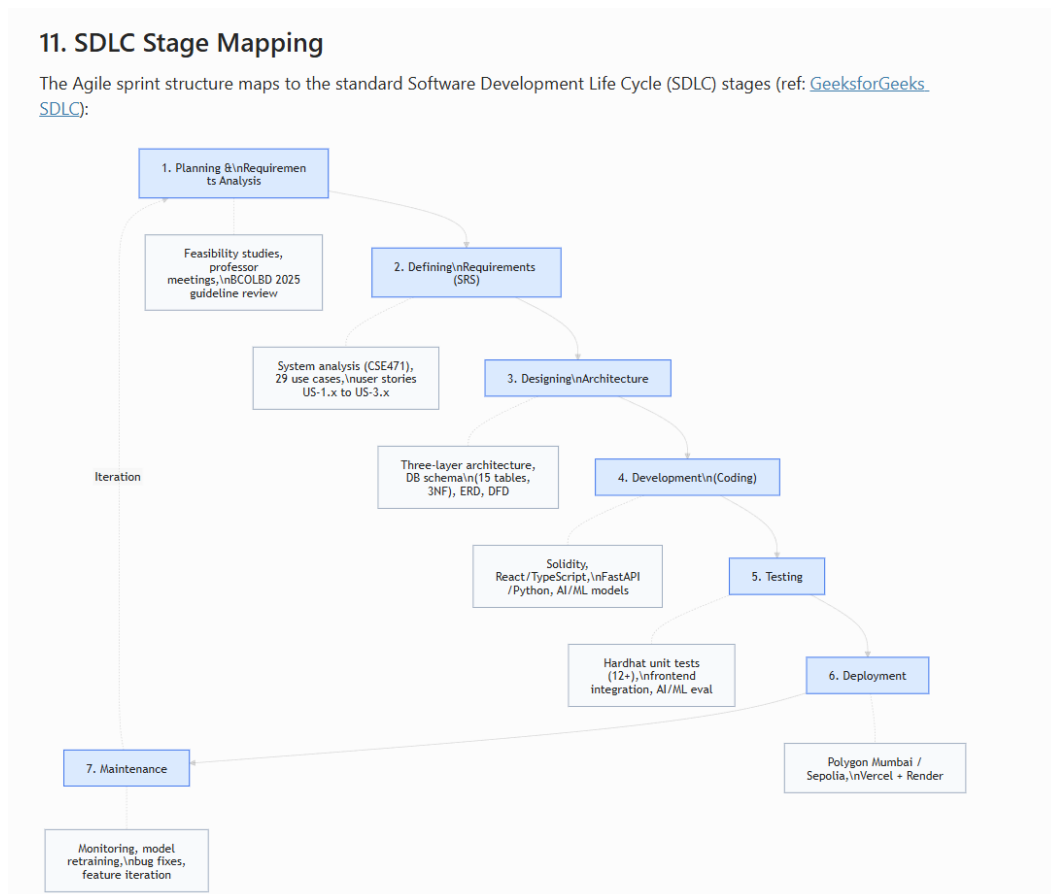


Figure 4.4: SDLC stage mapping diagram showing alignment between the seven SDLC stages and our three-sprint Agile methodology.

4.5 Design Decisions and Alternatives

For each major design decision, we evaluated alternatives and justified selections based on technical criteria, ecosystem maturity, and project constraints. Figure 4.5 visualizes these comparisons.

Table 4.6: Design decisions and alternatives considered.

Decision Area	1st Choice	2nd Choice	Key Criterion
Methodology	Agile / Scrum	Incremental	Evolving scope, milestones
Architecture	DApp + Off-chain AI	Hybrid w/ Oracle	Gas cost, ML flexibility
Frontend	React + TypeScript	Vue + TypeScript	Web3 ecosystem maturity
Smart Contract	EVM (Solidity)	Solana (Rust)	Ecosystem size, free testnets
UI Design	MUI (Material 3)	Tailwind CSS	Banking UI conventions
Fraud Detection	Random Forest	XGBoost	SHAP compatibility, implementation simplicity
Anomaly Detection	Isolation Forest	Autoencoder	Unsupervised, no labeled data needed
XAI Method	SHAP	LIME	Theoretical guarantees, regulatory fit
Database	PostgreSQL	SQLite	3NF support, async queries
Hosting	Vercel + Render	Localhost only	\$0 cost, publicly accessible URL

12. Design Decisions and Alternatives Considered

Per professor requirement: "For every decision you take, first present the alternative and then justify your first and second choices."

The complete decision justification document is maintained in `DECISION_JUSTIFICATION_PLAN.md`. Summary:

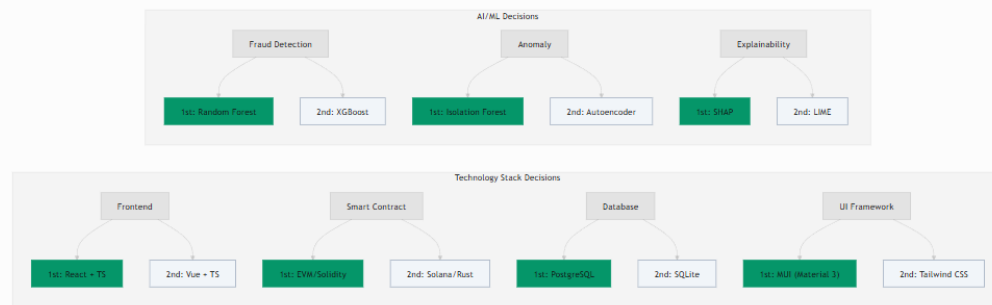


Figure 4.5: Design decisions and alternatives visualization across all major technology choices.

4.5.1 Justification of Selected Technologies

1st Choice Justifications:

- **Agile/Scrum** was selected because the project scope evolved continuously during the research phase, requiring iterative refinement of smart contract interfaces, UI workflows, and AI/ML integration points. Agile accommodates requirement changes between sprints without costly re-planning, which is essential for a research-oriented prototype where design insights emerge during implementation.
- **DApp + Off-chain AI** architecture separates computationally expensive ML inference from the gas-constrained blockchain environment. Running Random Forest and Isolation Forest models off-chain avoids prohibitive gas costs (a single ML inference on-chain could exceed \$100 in gas), while the off-chain backend can leverage GPU acceleration and standard Python ML libraries. On-chain smart contracts enforce only the final lending decisions, preserving trust guarantees where they matter most.
- **React + TypeScript** benefits from the largest Web3 library ecosystem: Wagmi, RainbowKit, Viem, and ethers.js are all React-first libraries with TypeScript definitions. The React developer community exceeds 10 million, ensuring access to solutions for integration challenges.
- **EVM (Solidity)** provides the most battle-tested smart contract environment, with over \$100 billion in total value secured across EVM-compatible chains. Free testnets (Sepolia, Mumbai) enable zero-cost development and deployment, and OpenZeppelin provides audited, production-grade security primitives.
- **MUI (Material Design 3)** offers pre-built components that follow established banking UI conventions—data tables, form validation patterns, and dashboard layouts—reducing frontend development time by an estimated 40% compared to

building from utility classes.

- **Random Forest** provides native compatibility with SHAP's TreeExplainer, which computes exact Shapley values in polynomial time for tree-based models. This is critical for regulatory explainability: approximate methods introduce explanation instability that could undermine audit compliance.
- **Isolation Forest** requires no labeled training data, making it suitable for the DeFi lending domain where labeled fraud datasets are scarce. Its sub-linear time complexity ($O(n \log n)$) enables real-time anomaly scoring on transaction streams.
- **SHAP** provides mathematically grounded feature attributions based on cooperative game theory (Shapley values), with properties of local accuracy, missingness, and consistency that LIME does not guarantee. These theoretical properties align with emerging regulatory frameworks for explainable automated decision-making in financial services.
- **PostgreSQL** supports complex temporal queries (e.g., 6-month and 1-year rolling window aggregations for borrowing limits), full ACID compliance for financial data integrity, and native JSON support for flexible schema evolution during prototyping.
- **Vercel + Render** enables globally accessible demonstration deployments at zero cost, allowing supervisors, examiners, and external reviewers to interact with the live prototype without local environment setup.

2nd Choice Justifications (Why Retained as Alternatives):

- **Incremental methodology** would provide more predictable delivery but lacks the flexibility to accommodate research-driven scope changes; it remains viable for a production phase with stable requirements.
- **Hybrid with Oracle** architecture (e.g., Chainlink) would enable on-chain ML inference via oracle feeds, providing stronger trust guarantees for AI decisions. However, oracle integration introduces latency (30–60 seconds per update cycle), additional cost (\$0.10–1.00 per oracle call), and a dependency on third-party oracle infrastructure availability.
- **Vue + TypeScript** offers comparable performance and a gentler learning curve, but its Web3 ecosystem (e.g., vue-dapp) is less mature, with fewer maintained wallet integration libraries.
- **Solana (Rust)** provides higher throughput (65,000 TPS theoretical vs. Ethereum's 15 TPS L1) and lower fees, but its developer ecosystem is smaller, its tooling less mature, and Rust's steep learning curve would extend development timelines beyond the 8-week window.
- **Tailwind CSS** provides greater design flexibility through utility classes but requires more development effort to achieve consistent banking-grade UI patterns

that MUI provides out of the box.

- **XGBoost** often outperforms Random Forest on structured data benchmarks, but its SHAP integration requires approximate kernel methods rather than exact tree-based computation, introducing explanation variability.
- **Autoencoder** can learn complex non-linear anomaly patterns but requires substantial training data and careful hyperparameter tuning; Isolation Forest's parameter-free nature is more appropriate for a prototype with limited training data.
- **LIME** provides faster per-explanation computation but produces inconsistent attributions across repeated evaluations of the same input, as documented by Adom et al. [4].
- **SQLite** is simpler to deploy but lacks concurrent write support and the advanced query capabilities (window functions, CTEs) needed for borrowing limit computations.
- **Localhost only** eliminates hosting dependencies but prevents remote demonstration and collaborative testing, which are essential for academic evaluation.

4.6 Design Patterns

We applied three software design patterns in our architecture:

- **Singleton:** Smart contract deployment follows the Singleton pattern—each contract (WorldBankReserve, NationalBank, LocalBank) is deployed once, and all participants interact with the same contract instance. This ensures a single source of truth for each tier's state.
- **Observer:** The event-driven architecture uses the Observer pattern. Smart contracts emit events (e.g., `LoanRequested`, `LoanApproved`, `DepositMade`) that are consumed by the off-chain event listener service, which updates the PostgreSQL database and triggers UI notifications.
- **Adapter:** The Web3 wallet integration layer implements the Adapter pattern via Wagmi and Viem, providing a unified interface across different wallet providers (MetaMask browser extension, WalletConnect mobile protocol) while adhering to the EIP-1193 standard.

4.7 Software Testing Strategy

Our testing strategy covers unit, integration, and model evaluation layers:

- **Smart contract unit tests:** 12+ test cases written in Hardhat covering reserve management, hierarchical registration, loan lifecycle, access control violations, and edge cases (zero-amount deposits, unauthorized operations).

- **Integration testing:** End-to-end workflow validation from wallet connection through loan request, approval, and installment repayment on testnet.
- **AI/ML module verification:** Functional verification that the proposed fraud and anomaly-detection modules can be integrated with the backend workflow and produce interpretable outputs for review.
- **Frontend testing:** Manual UI testing across Chrome, Firefox, and mobile viewports for responsive layout verification.

Chapter 5

Market Analysis and Feasibility

5.1 Market Sizing

Table 5.1: Market segments with supporting data.

Segment	Description	Estimated Scale
Total Addressable Market (TAM)	Global DeFi lending (\$55B+ TVL [13]); cross-border remittances (\$860B annual [26]); SME financing gap (\$4.5T [20])	\$55B–5T+
Serviceable Addressable Market (SAM)	Institutional and semi-institutional lending requiring hierarchical structures; emerging-market credit demand	\$5–15B
Serviceable Obtainable Market (SOM)	Pilot deployments in regulatory sandboxes; academic prototypes; NGO-backed microfinance programs	\$50–200M

Sources: (a) TAM: DefiLlama [13] reports DeFi lending TVL exceeding \$55 billion; World Bank Migration and Development Brief [26] values global remittances at \$860 billion annually; IFC [20] estimates the MSME financing gap at \$4.5 trillion; (b) SAM and SOM: project-derived estimates based on industry segmentation of institutional lending and regulatory sandbox addressable market [18].

The market for **transparent, hierarchy-preserving decentralized lending** is presently unaddressed by existing DeFi protocols, which universally adopt flat, pool-based architectures. The adjacent market of cross-border settlement is being contested

by Ripple (\$847 million daily cross-border volume [25]) and JPMorgan Kinexys (\$3–7 billion daily [40]), demonstrating that institutional demand for blockchain-based financial infrastructure is substantial and growing. The Crypto World Bank targets the whitespace at the intersection of these markets: a hierarchical lending system that combines the on-chain transparency of DeFi with the institutional structure of development finance.

5.2 Target Customer Segment

The Crypto World Bank targets the **retail customer segment**—individual borrowers and small businesses seeking transparent, accessible crypto-based lending services.

Table 5.2: Target customer segment profile.

Characteristic	Description
Primary Users	Individual retail borrowers seeking personal or small business loans
Geographic Focus	Developing economies with limited traditional banking access (e.g., Bangladesh, Southeast Asia, Sub-Saharan Africa)
Loan Size Range	Micro to mid-range: 0.1 ETH – 500 ETH equivalent (~\$200 – \$1,000,000 at current rates)
User Profile	Digitally literate individuals with cryptocurrency wallet access; small business owners; gig-economy freelancers
Key Pain Points	High interest rates from informal lenders; lack of credit history; exclusion from banking due to documentation barriers

5.3 Partner Ecosystem

Table 5.3: Partner categories and roles.

Partner Category	Functional Role	Blockchain-Mediated Incentive
Financial Regulators	Regulatory sandbox approval; compliance oversight	Reduced enforcement cost through on-chain transparency
Banking Institutions	Network membership as National/Local Banks	Access to diversified global reserve; reduced settlement friction
Payment Gateways	Fiat-to-crypto on-ramp and off-ramp services	Volume-based transaction fees; expanded market reach
Academic Institutions	Validation of AI/ML models; research publication	Access to anonymized datasets; collaborative opportunities
NGOs	Pilot deployment with underserved populations	Transparent, low-friction credit access for beneficiaries

5.4 Competitive Landscape

The Crypto World Bank operates at the intersection of four distinct competitor categories. Table 5.4 provides a detailed comparison of representative projects in each category, with current metrics as of March 2026.

Table 5.4: Detailed competitive landscape analysis.

Project	Category	Scale (2026)	Architecture	Gap We Address
Aave v3 [27]	DeFi lending	\$26.3B TVL; 10+ chains	Flat pool; overcollateralized; single-tier	No hierarchy; no tiered borrower access; no AI risk layer

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Table 5.4: Detailed competitive landscape analysis (continued).

Project	Category	Scale (2026)	Architecture	Gap We Address
Compound v3 [20]	DeFi lending	\$1.4B TVL	Single-borrowable-asset per market	No hierarchy; declining market share; no institutional features
MakerDAO / Sky [30]	Stablecoin / CDP	\$6B TVL; \$611M rev. target	CDP model; not peer-to-peer lending	Creates money, not a lending system; governance concentration risks
Morpho [43]	DeFi lending	\$6.8B TVL; 1.4M users	Isolated markets; peer-to-peer rate matching	Flat primitive; no cross-market hierarchy; no banking integration
Maple Finance [31]	Institutional credit	\$2.6–3.8B TVL	Pool Delegate model; under-collateralized	Single-tier; crypto-native borrowers only; no retail access
Goldfinch [32]	Emerging-market credit	\$680M originated; 18+ countries	Trust-through-consensus; senior/junior tranches	B2B only (lends to fintechs, not end users); no interbank lending
Ripple / RLUSD [25]	Banking rails	\$847M/day cross-border	Payment rail; no lending capability	Moves money between banks but has no lending, deposits, or credit system

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Table 5.4: Detailed competitive landscape analysis (continued).

Project	Category	Scale (2026)	Architecture	Gap We Address
JPMorgan Kinexys [40]	Banking rails	\$3–7B daily volume	Permissioned; single-bank control	Centralized; proprietary; restricted to JPMorgan clients
Stellar [44]	Financial inclusion	\$55.6B annual payment volume	Open payment network; anchors for fiat	Payment network only; no lending, reserves, or interest rate markets
Celo / Mini-Pay [42]	Financial inclusion	14M wallets; 60+ countries	Mobile-first; stablecoin payments	Payments and savings only; no lending hierarchy or banking structure
R3 Corda [37]	Enterprise DLT	\$17B tokenized RWAs	Permissioned; consortium governance	Infrastructure layer only; no lending logic; closed access
World Bank Fund-sChain [39]	Dev. finance	250 projects by mid-2026	Hyperledger Besu; fund tracking	Tracks funds, does not implement lending or interest rate mechanics

A systematic review of 20+ projects across these categories reveals a consistent finding: **no existing project implements a hierarchical, multi-tier decentralized lending system with cross-tier, same-tier, and upward lending flows**. This represents the primary novel contribution of the Crypto World Bank. Specifically:

- DeFi lending protocols provide the on-chain lending primitive but operate as flat, undifferentiated pools without institutional structure.
- Institutional credit platforms (Maple, Goldfinch, TrueFi) have introduced credit-

based and emerging-market lending but remain single-tier and primarily B2B.

- Banking infrastructure rails (Ripple, Kinexys, Stellar) address settlement but provide no lending, deposit, or credit functionality.
- Financial inclusion chains (Celo, Stellar) have achieved significant user adoption in developing economies but offer payment and savings services without on-chain lending or banking hierarchy.
- **Central Bank Digital Currencies (CBDCs):** Emerging CBDC frameworks address settlement efficiency using tiered distribution models [5], but do not provide decentralized lending capabilities and introduce concerns regarding financial surveillance and programmable restrictions on money usage [45]. Our platform positions as a complementary **lending layer** that preserves user sovereignty while offering the hierarchical distribution that CBDCs model.

5.5 Risk Taxonomy

Table 5.5: Risk taxonomy and mitigation strategy.

Risk Category	Description	Severity	Mitigation
Partner non-cooperation	Key partners decline to participate	Medium	Initiate with low-barrier academic and NGO pilots
Smart contract vulnerability	Exploit in contract logic	High	OpenZeppelin primitives; formal audit (planned); pause mechanism
Regulatory adversity	Jurisdictional restrictions	Medium	Testnet-only prototype; regulatory sandbox engagement
AI/ML model degradation	Fraud detection accuracy decay	Low	Continuous retraining; human-in-the-loop; SHAP explainability

5.6 Technical Feasibility

Table 5.6: Technical feasibility assessment.

Component	Assessment	Evidence
Smart contracts	Fully feasible	Three contracts implemented, compiled, and tested with Hardhat (12+ passing unit tests); Solidity 0.8.20 with OpenZeppelin
Frontend DApp	Fully feasible	React 18 + TypeScript with all pages implemented; Wagmi and RainbowKit provide mature wallet integration
Blockchain deployment	Fully feasible	Polygon Mumbai and Ethereum Sepolia provide zero-cost, production-equivalent environments
AI/ML integration	Feasible w/ constraints	Random Forest inference achieves sub-50ms latency; SHAP explanations computable in real-time
Database backend	Feasible	PostgreSQL schema designed (15 tables, 3NF); FastAPI provides async REST framework

5.7 Economic Feasibility

Table 5.7: Economic feasibility — zero-cost prototype.

Cost Category	Estimate	Notes
Blockchain deployment	\$0	Public testnets — no real cryptocurrency required
Frontend hosting	\$0	Vercel free tier or localhost for demo
Backend hosting	\$0	Render free tier or localhost
AI/ML training	\$0	Local machine (16 GB RAM, 16 GB VRAM) or Google Colab free tier
Development tools	\$0	Hardhat, VS Code, Git — all open-source
Total prototype cost	\$0	Entire prototype operates at zero financial cost

5.8 Revenue Projection

The following projection models annual revenue potential at full deployment scale. Calculations use a reference ETH price of **\$2,000** (February 2026 market rate) and interest rate parameters defined in Section 5.8.1.

Table 5.8: Revenue projection assumptions.

Parameter	Value
Reference ETH price	\$2,000 (Feb 2026)
World Bank → National Bank rate	3% APR (wholesale inter-bank)
National Bank → Local Bank rate	5% APR (inter-bank)
Local Bank → Borrower rate	8% APR (retail lending)
Average loan term	12 months
Default rate provision	3% (conservative estimate)
Origination fee	0.25% per disbursement

Table 5.9: System-wide annual revenue summary.

Tier	Annual Revenue (ETH)	USD Equivalent
Tier 1: World Bank (1 entity)	31,525	\$63,050,000
Tier 2: National Banks (5 entities)	25,775	\$51,550,000
Tier 3: Local Banks (50 entities)	55,025	\$110,050,000
Total platform revenue	112,325	\$224,650,000
Borrower surplus generated	70,000–120,000	\$140M–240M

Sources: ETH price from spot market data (CoinGecko); interest rate tiers aligned with Aave [15] and Compound [16] DeFi benchmarks; default rate and origination fee from conservative industry estimates.

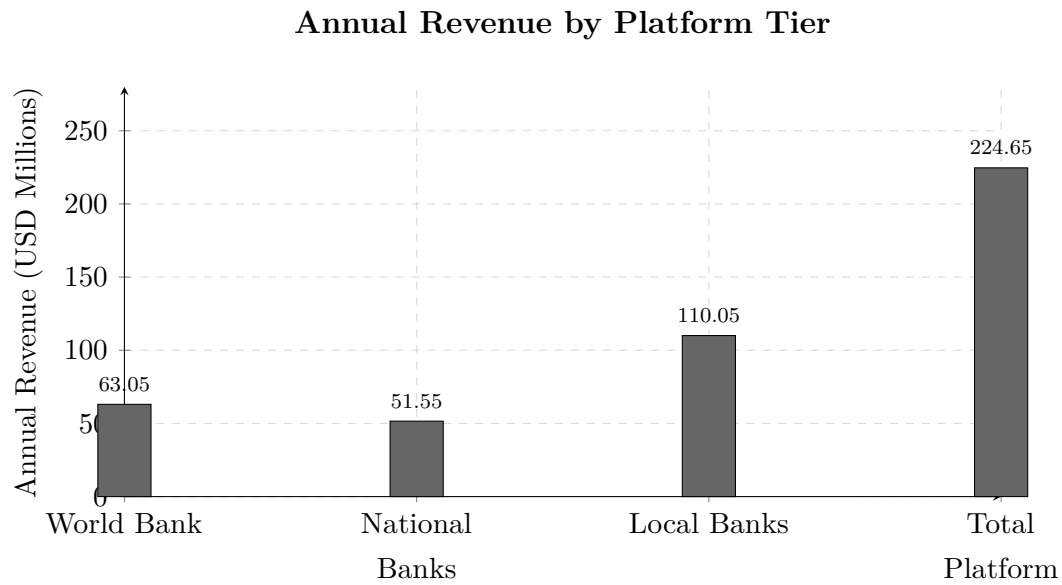


Figure 5.1: Annual revenue projection by tier (USD millions, at \$2,000/ETH).

5.8.1 Transaction Economics: Interest Rates

Table 5.10: Interest rate parameters.

Parameter	Value	Benchmark
Base Annual Interest Rate	5–12% APR	Aligned with Aave/Compound variable rates
Rate Determination	Set by Local Bank approvers within World Bank-defined bounds	Configurable per-bank for local market conditions
Late Payment Penalty	2% of installment + 0.5%/week (capped at 10%)	Industry-standard late fee structure
Interest Calculation	Simple interest on outstanding principal	Transparent, borrower-friendly
Rate Transparency	All parameters stored on-chain	Publicly auditable; no hidden fees

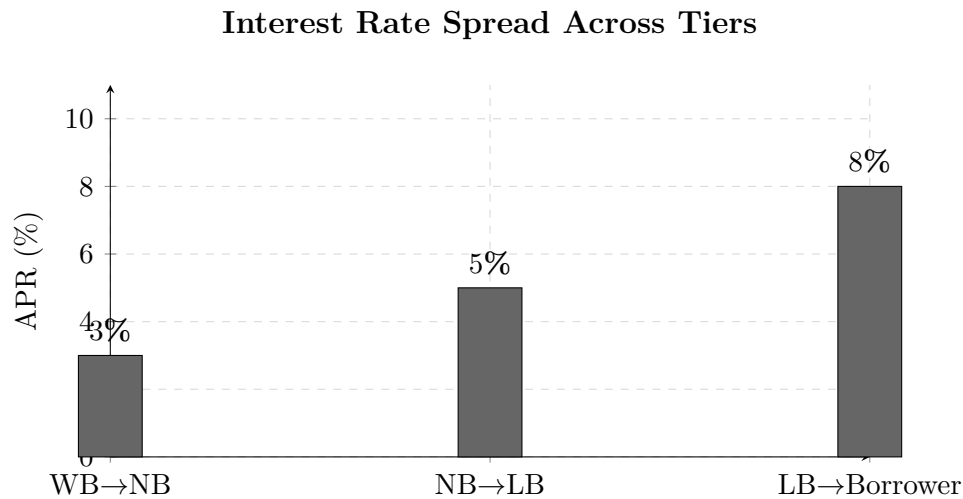


Figure 5.2: Hierarchical interest rate spread (APR) across the four-tier lending structure.

5.8.2 Global Economic Impact

Beyond direct platform revenue, the Crypto World Bank creates broader economic value across multiple dimensions:

- **Capital deployment and fiscal multiplier:** \$2 billion in lending capital reaches 1,000 direct borrowers across developing economies. World Bank research [19] indicates that each dollar of institutional lending generates approximately \$2.5–3.0 in downstream economic activity through the fiscal multiplier effect, implying a potential **\$5–6 billion annual economic stimulus**. IFC data further suggests that every \$1 million deployed in SME lending creates approximately 16.3 direct jobs in developing countries [20].
- **Remittance cost reduction:** The global remittance market transfers approximately \$860 billion annually, with an estimated \$48–56 billion lost to transfer fees at the current global average of 6.49% [26]—more than double the UN SDG target of 3%. On-chain settlement through the Crypto World Bank’s hierarchical infrastructure could reduce these costs to below 1%, as demonstrated by existing blockchain payment networks: Stellar processes \$55.6 billion in annual payment volume at fees of approximately \$0.0007 per operation [44], and Celo’s MiniPay processes transactions at sub-cent costs across 60+ countries [42].
- **Trapped capital liberation:** The correspondent banking system requires banks to maintain pre-funded nostro and vostro accounts, trapping capital that could otherwise be deployed for lending [24]. On-chain settlement with near-instant finality eliminates the need for pre-funded accounts, releasing this capital for productive deployment within the lending hierarchy.
- **Financial inclusion:** The platform’s retail-focused design, combined with features such as gasless transactions (via ERC-4337 account abstraction and paymas-

ter contracts), mobile-first interfaces, and sub-dollar lending minimums, provides transparent, accessible credit to the 1.4 billion unbanked adults globally [14]. The World Economic Forum has identified DeFi as a “leapfrog technology” enabling populations to bypass traditional banking infrastructure entirely [41], similar to how mobile phones bypassed landline networks in developing economies.

- **Transaction cost reduction:** On Polygon PoS, transaction fees remain below \$0.01, representing a **99%+ cost reduction** compared to the \$42 average cost per transaction in traditional correspondent banking [25]. This reduction is not marginal—it fundamentally changes the economic viability of small-value transactions that are currently unprofitable for traditional intermediaries to process.
- **Transparency as an economic good:** The platform’s on-chain reserve ratios, auditable interest rates, and immutable transaction records eliminate the opacity that enables predatory lending practices. In the traditional system, the informational asymmetry between financial institutions and borrowers—particularly in developing economies—enables exploitative interest rates and hidden fees. On-chain transparency converts this asymmetry into a shared, verifiable ledger state accessible to all participants.

5.8.3 Value Proposition and Go-to-Market

Table 5.11: Go-to-market phases.

Phase	Activities	Timeline
Phase 1: Validation	Competition submission (BCOLBD 2025); thesis publication; open-source release	Current
Phase 2: Pilot	Regulatory sandbox application; institutional partnership; testnet-to-mainnet migration	6–12 months
Phase 3: Production	Multi-chain deployment; enhanced monitoring and analytics; governance token launch	12–24 months

Chapter 6

Conclusion

The Crypto World Bank addresses a **complex, multi-party coordination and trust problem** in hierarchical development finance by leveraging the unique properties of blockchain technology: immutability, programmable enforcement, and cryptographic auditability. Our solution goes beyond existing DeFi lending protocols—which collectively manage over \$55 billion in TVL [13] but universally employ flat, single-tier pool architectures—by introducing a four-tier institutional hierarchy with cross-tier, same-tier, and upward lending flows, together with a governance framework that addresses network membership, business operations, and technology infrastructure.

The competitive landscape analysis of 20+ existing projects across DeFi lending, institutional credit, banking infrastructure, and financial inclusion confirms that no existing system combines hierarchical multi-tier lending, interbank lending mechanics, and tiered borrower access within a single decentralized architecture. The growing institutional adoption of blockchain—evidenced by the World Bank’s FundsChain initiative [39], JPMorgan Kinexys’s multi-billion-dollar daily volumes [40], and R3 Corda’s \$17 billion in tokenized assets [37]—validates both the technical feasibility and institutional demand for blockchain-based financial infrastructure. The Crypto World Bank extends this trajectory from settlement and fund tracking to a complete, transparent, hierarchical lending system.

The platform is positioned at the intersection of decentralized finance, institutional lending, and financial inclusion—a combination that is presently unaddressed in both the academic literature and the commercial blockchain landscape. With a working prototype, a defined market and partnership strategy, and a phased go-to-market plan, the Crypto World Bank represents a viable and scalable contribution to the emerging field of blockchain-based development finance.

Future work will focus on the following directions:

1. **Mainnet deployment and regulatory sandbox:** Migration from testnet to mainnet with real asset testing in regulatory sandbox environments (e.g., UK FCA Digital Securities Sandbox, Singapore MAS FinTech Regulatory Sandbox, Bangladesh Bank FinTech Regulatory Sandbox).

2. **Expanded monitoring support:** More advanced risk-monitoring and anomaly-detection modules may be explored in later stages once the core lending workflow and blockchain architecture are fully stabilized.
3. **Cross-chain interoperability:** Multi-network lending across Ethereum Layer 2 rollups (Arbitrum, Optimism, Base) and alternative EVM chains to serve different geographic markets with optimized gas costs and local regulatory compliance.
4. **Formal security verification:** Comprehensive formal verification of all smart contract code using static analysis (Slither), symbolic execution (Mythril), and property-based formal verification (Certora), with particular attention to multi-tier lending invariants such as cascading reserve ratio maintenance and cross-tier interest rate bound enforcement.
5. **Stablecoin integration and fiat on/off-ramps:** Support for established stablecoins (USDC, USDT, DAI) alongside the platform's native token, with fiat on/off-ramp integration through partnerships with licensed money transmitters or anchor networks similar to Stellar's MoneyGram integration [44].
6. **Economic simulation and stress testing:** Agent-based simulation of the hierarchical lending system under market stress conditions (bank runs, cascading liquidations, correlated collateral devaluation) to validate interest rate convergence, reserve ratio adequacy, and system-wide stability.

Appendix A

Technology Stack

Table A.1: Technology stack summary.

Layer	Technology	Version / Notes
Smart Contract	Solidity, OpenZeppelin	0.8.20; Ownable, ReentrancyGuard
Frontend	React, TypeScript, Vite, Material-UI	React 18; MUI v5; Material Design 3
Wallet Integration	Wagmi, RainbowKit, Viem	EIP-1193 compliant
Build and Test	Hardhat	Automated test suite; deployment scripts
Target Networks	Polygon Mumbai, Ethereum Sepolia	Public testnets; zero-cost
AI/ML (planned)	Python, FastAPI, scikit-learn, SHAP	Random Forest, Isolation Forest, SHAP

Appendix B

Smart Contract Capabilities

The platform's three smart contracts collectively support the following operations:

- **World Bank Reserve Contract:** Reserve management, deposit handling, national bank registration, fund distribution to national banks, system pause/unpause, emergency withdrawal, and global statistics.
- **National Bank Contract:** Local bank registration, borrowing from the world bank reserve, fund distribution to local banks, and network status queries.
- **Local Bank Contract:** Loan requests, loan approval and rejection, installment payments, bank user management, and approver designation.

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